

**DEVELOPMENT OF A HOST BASED
INTRUSION DETECTION SYSTEM USING
RANDOM FOREST FOR A HI TECH VAULT**

BY

AGALI EMMANUEL CHUKWUDI

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**A PROJECT SUBMITTED TO
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DECLARATION

I, Agali, Emmanuel Chukwudi, a B.Sc. Cyber Security student in the Department of Mathematical and Computing Sciences, KolaDaisi University, Ibadan, hereby declare that this project titled “ Development of an Intrusion detection system using Random forest hi tech vault”, submitted by me is based on my original work. Any material(s) obtained from other sources or work done by any other persons or institutions have been duly acknowledged.

Student's Full Name and Matriculation number

Signature & Date

CERTIFICATION

This is to certify that this Project has been read and approved as meeting the requirements of the Department of Mathematical and Computing Sciences, KolaDaisi University, Ibadan, Nigeria for the award of Bachelor of Science (B.Sc.) degree in Cyber Security.

.....

Dr. Oyediran Mayowa

Supervisor

.....

DATE

.....

Mr Korede Soloman Olagunju

Co- Supervisor

.....

DATE

.....

Dr. Latifat A. Odeniyi

Head of Department

.....

DATE

DEDICATION

I dedicate this project to God, for his unwavering guidance and mercy and never changing grace throughout my journey.

I also dedicate this project to my loving Mother; Miss Elizabeth Ajudua, my Pastor; Pastor Owen Nlekwuwa for their consistent love and support all through my years in school. Their encouragement has been my driving force inspiring me to persevere and strive for excellence in completing this project.

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ABSTRACT

The results highlight how well sensor-level monitoring and machine learning work together to improve host-based security, especially in settings that demand adaptive, real-time threat detection. This study offers a scalable framework that can be applied to other crucial security infrastructures and adds to the expanding corpus of knowledge on intelligent intrusion detection. In critical environments like high-tech vaults, the growing convergence of cyber-physical systems calls for sophisticated security measures that can identify both digital and physical intrusions. The creation of a host-based intrusion detection system (HIDS) that can monitor and react to unusual activity at the device level is presented in this study. The suggested system combines a number of physical sensors, such as RFID modules, temperature sensors, and gyroscopes, and interfaces them with a Raspberry Pi. Data is centrally stored in a MySQL database.

Sensor data was analyzed and events were classified as normal or anomalous using a Random Forest machine learning algorithm. According to experimental results, the system performed well in detecting malicious activity while reducing false positives, achieving an overall accuracy of 99%, a precision of 1.00, and an F1-score of 0.93 for anomaly detection. The results highlight how well sensor-level monitoring and machine learning work together to improve host-based security, especially in settings that demand adaptive, real-time threat detection. This study offers a scalable framework that can be applied to other crucial security infrastructures and adds to the expanding corpus of knowledge on intelligent intrusion detection.

CHAPTER 1

INTRODUCTION

1.0. Background to the Study

This project focuses on the implementation and development of a Host-based Intrusion Detection System (HIDS) specifically tailored for a Hi-tech vault system, leveraging the power of a machine learning algorithm known as Random Forest. The HIDS system is designed to monitor and analyze data from various environmental and access sensors embedded within the vault's host systems in real-time, aiming to detect any form of abnormal behavior or malicious activities that indicate a physical or cyber-physical intrusion.

Due to the rapid advancement of technology, new and increasingly sophisticated attack vectors are constantly being presented by malicious actors and organized criminal groups. Cyber security has become a paramount concern within the cyberspace, affecting individuals, organizations, and national critical infrastructures alike (Sampath & Anuradha, 2023). The pervasive nature of computer systems and networked environments has consequently elevated the need for robust protection mechanisms, driving continuous innovation in security solutions. As network usage proliferates, so too do intrusion activities, making the development of effective and efficient intrusion detection systems (IDS) a critical priority in recent years (Kumar et al., 2019).

When it comes to safeguarding both digital and physical assets, particularly within high-security environments such as a Hi-tech vault system, security stands as one of the most critical concerns. An Intrusion Detection System (IDS) is fundamentally a software application or a combination of hardware and software that monitors system or network activities for suspicious patterns and raises an alarm when anomalous or malicious behavior is detected (Yogesha & Reddy, 2022). The foundational concept of intrusion detection was pioneered by James P. Anderson in his seminal 1980 report, "Computer Security Threat Monitoring and Surveillance," which laid the groundwork for today's sophisticated IDS solutions (Farnaaz & Jabbar, 2016). Intrusion Detection Systems are broadly categorized into two major types: Network-based Intrusion Detection Systems (NIDS) and Host-based Intrusion Detection Systems (HIDS). While NIDS monitors network traffic for suspicious packets traversing a network segment, a Host-based Intrusion Detection System (HIDS)

operates on individual hosts, servers, or in this case, the specific computing components managing a physical vault system. A HIDS scrutinizes local system activities such as file accesses, system calls, process executions, and crucially, security logs generated by integrated physical sensors and access control mechanisms (Farnaaz & Jabbar, 2016). For the unique context of a hi-tech physical vault, HIDS is particularly relevant as it provides granular visibility into the integrity and behavior of the embedded sensors and control systems, which is essential for detecting physical tampering or unauthorized access attempts.

The protection of highly valuable items within a hi-tech vault system has been evolving, but traditional surveillance systems (e.g., CCTV, basic alarm systems) and static access control systems (e.g., keypads, card readers) are increasingly proving to be outdated and insufficient against modern threats (Kumar et al., 2019). New and emerging threats, including sophisticated attempts at sensor manipulation, environmental control circumvention, and stealthy physical breaches, require intelligent systems capable of proactively analyzing, identifying, preventing, and escalating any form of security breach before significant damage can be incurred. Therefore, an intrusion detection system leveraging machine learning algorithms offers a much more dynamic, adaptive, and intelligent solution to this evolving security challenge compared to rigid rule-based systems (Farhan et al., 2022).

In today's era of rapid technological development and pervasive connectivity, cybersecurity presents an increasingly complex problem. With humanity's vast knowledge accessible through the internet and the widespread adoption of mobile computing, cybercrime and cyber-attacks have become alarmingly common. A statement from the Anti-Phishing Working Group (APWG) indicates that around 227,000 virus detections occur every day, with approximately 20 million new malwares discovered daily (Yogesha & Reddy, 2022). A malicious computer program created to target a system is broadly known as malware. While malware was historically easier to contain, the landscape of cybercrime has significantly intensified over the past two decades, prompting the evolution of cybersecurity tactics into more sophisticated, data-driven methods. The challenge extends beyond software vulnerabilities; it encompasses the requirement to protect physical systems from cyber-physical attacks where digital means are used to compromise physical security (Sampath & Anuradha, 2023).

A major problem that has arisen as a result of technological advancements and the proliferation of networks is the increasing ease with which even inexperienced criminals can launch sophisticated cyber-attacks. The availability of advanced programs and various scripts allows attackers to bypass and avoid security measures. According to an APWG analysis from 2016, phishing attempts alone cost billions of dollars, and the retail industry has been the target of several large-scale attacks (Yogesha & Reddy, 2022). Intrusion detection systems play a crucial role in overseeing a large volume of information and data generated by system activities, and are critical in identifying many types of intrusions. Fundamentally, IDS is a classification challenge, as it depends on a robust classifier to distinguish between normal and malicious activities (Yogesha & Reddy, 2022; Kumar et al., 2019). A damaging act that compromises a computer system, whether digitally or physically, is defined as an intrusion. Consequently, due to the persistent threat of network and physical attacks, anomaly and intrusion detection systems have grown profoundly in importance for comprehensive system defense (Sampath & Anuradha, 2023).

Random Forest algorithm is a highly effective ensemble classification and regression approach, widely recognized as one of the most powerful techniques for data mining and machine learning (Yogesha & Reddy, 2022). It is chosen for its superior accuracy, reliability, and its inherent ability to handle complex datasets with multiple features, including both numerical and categorical data. Conceptually, Random Forest constructs a multitude of decision trees during training and outputs the class that is the mode of the classes (for classification) or mean prediction (for regression) of the individual trees (Yogesha & Reddy, 2022). This ensemble approach mitigates the overfitting issues commonly associated with single decision trees and significantly enhances predictive power and robustness.

This project focuses on creating a Host-based Intrusion Detection System that is not only capable of detecting intrusions in real-time but also possesses adaptive capabilities to evolve with new threats by continuously learning from the real-time data acquired from various sensors within the vault environment.

While the Random Forest algorithm has been used extensively in diverse applications, including predictive modeling, medical diagnosis, financial fraud detection (Yogesha & Reddy, 2022), and significantly, in network intrusion detection systems (NIDS), where it has shown high accuracy and low false alarm rates in classifying various types of network attacks (Kumar et al., 2019;

Farhan et al., 2022; Farnaaz & Jabbar, 2016), its specific application and rigorous evaluation within a Host-based Intrusion Detection System (HIDS) tailored for the unique sensor profiles and physical environment of a hi-tech vault system presents a novel research opportunity. This project specifically seeks to bridge the gap by applying this robust algorithm to physical security through the analysis of diverse sensor data, thereby differentiating it from existing network-centric approaches and emphasizing its HIDS nature.

A hi-tech vault system is a contemporary, sensor-enabled, and highly secure storage space that integrates advanced technologies such as real-time alerts, sophisticated environmental monitoring (e.g., temperature, humidity, vibration, pressure), and digital access control mechanisms (e.g., biometrics, smart card readers, multi-factor authentication systems). Unlike rudimentary physical strongboxes, these systems are designed to be proactive and intelligent. Traditional surveillance and manual monitoring systems are no longer adequate due to the growing sophistication and stealth of modern intrusion attempts, which can involve subtle environmental changes or attempts to bypass individual sensors. Because of this, there is an urgent need for intelligent systems that can learn and adjust to new types of intrusion attempts, identifying subtle anomalies that human operators or static rule sets might miss. The integration of an intelligent Host-based Intrusion Detection System (HIDS), capable of processing these varied sensor inputs, is therefore paramount for next-generation vault security.

1.1. Statement of the Problem

Most intrusion detection systems focus on Network-Based Intrusion Detection Systems (NIDS), which primarily monitor external traffic and data packets across a network. While this is effective for identifying threats targeting network infrastructures, it leaves a critical gap in the protection of individual devices and physical systems especially in environments like high-tech vaults where localized, sensor-level threats may not traverse traditional networks.

Most current HIDS implementations are limited, particularly in physical security applications. Most existing systems rely on static rule-based logic, which lacks adaptability to emerging threats. As a result, there is an urgent need for intelligent, adaptive host-level detection systems that can protect critical physical environments through proactive analysis and fast response mechanisms.

1.2. Aim and Objectives of the Study

1.2.1 Aim

The aim of this project is to develop a Host-based Intrusion Detection System (HIDS) using the Random Forest machine learning algorithm to enhance the security of a hi-tech vault system by monitoring its internal environment and access points.

1.2.2 Specific Objectives

The specific objectives are to:

- I. To develop a host-based Intrusion Detection model using the Random Forest algorithm for anomaly detection based on sensor data in a hi-tech vault environment.
- II. To implement the developed model in I. on a Hi tech vault device to monitor and analyze data streams from various internal vault sensors (e.g., motion, vibration, temperature, humidity, pressure, access logs) to detect any form of abnormal behavior or malicious activities in real-time.
- III. To evaluate the performance of the developed Host-based Intrusion Detection model using standard metrics including accuracy, detection rate, and false positive rate using simulated or real intrusion scenarios.

1.3. Scope of the Study

The scope of this study is limited to the development and evaluation of an intrusion detection system specifically designed for a high-tech vault environment. The system will primarily utilize the Random Forest machine learning algorithm for its detection capabilities. The system's training and testing will be based on either real-world or carefully simulated sensor and access data, designed to mimic typical vault operations and various intrusion attempts, including unauthorized access, environmental anomalies, and sensor tampering. The focus is strictly on the detection and prevention (through alerting) of intrusions, rather than on physical response mechanisms (e.g., automatically locking doors or deploying physical barriers). The system's performance will be rigorously evaluated based on standard performance metrics such as accuracy, precision, recall, and F-score, derived from controlled testing scenarios. While the project aims for practical applicability, the integration with actual physical hardware may be simulated or partially

implemented, depending on the availability of resources and equipment, with a clear focus on the software-based detection logic and its ability to act as a Host-based IDS.

1.4. Significance of the Study

This study is significant for several compelling reasons. It demonstrates a novel and practical application of advanced machine learning techniques, specifically Random Forest, within the critical domain of physical security systems. This bridges the gap between theoretical AI models and tangible, real-world asset protection, showcasing how intelligent algorithms, as part of a Host-based IDS, can enhance traditional security measures by providing proactive, data-driven insights from physical sensors.

The project thus contributes directly to the development of more secure, intelligent, and adaptive vault systems. By moving beyond static defenses and leveraging real-time sensor data analysis through a HIDS, the proposed system offers a dynamic approach to safeguarding high-value assets against an evolving threat landscape, significantly reducing the vulnerability to sophisticated intrusion methods and improving overall security posture.

The project provides a robust framework and methodology for implementing Host-based Intrusion Detection Systems in sensitive physical environments where immediate and accurate anomaly detection based on environmental and access data is paramount. This framework can serve as a blueprint for similar applications in other critical infrastructure, high-security facilities, or industrial control systems where continuous monitoring of physical parameters is vital.

The project also offers valuable insights into the efficacy and practical considerations of using the Random Forest algorithm for real-time classification problems in security contexts, particularly when dealing with diverse sensor data streams from a physical environment. The findings can inform future research and development in this specialized area, demonstrating Random Forest's adaptability beyond traditional network-centric IDS to encompass host-level physical monitoring.

The project lays a crucial foundation for future research and development in adaptive security systems that combine real-time hardware sensor data with sophisticated Artificial Intelligence models. This paves the way for a new generation of proactive and self-learning security solutions that can continuously improve their detection capabilities as new threat patterns emerge in both digital and physical realms.

CHAPTER TWO

LITERATURE REVIEW

2.0 overview

Intrusion Detection Systems (IDS) are critical cyber security mechanisms that monitor and analyze system or network activities to detect unauthorized access or anomalies. IDS can be broadly categorized into Host-Based Intrusion Detection Systems (HIDS) and Network-Based Intrusion Detection Systems (NIDS). While NIDS monitors network traffic, HIDS focuses on monitoring individual systems—making it particularly suitable for securing high-tech physical environments like vault systems (Yogesha et al., 2022).

Security mechanisms traditionally employed in vault systems possess limited capabilities in effectively detecting and responding to novel and sophisticated intrusion attempts. These conventional systems frequently rely on manual human monitoring, static predefined rules, and often operate as disparate systems that do not communicate efficiently or effectively (Kumar et al., 2019; Farhan et al., 2022). This fragmented approach typically results in several critical drawbacks: high false alarm rates, significantly slow response times, and inherent vulnerability to new, intelligent threats that bypass known signatures or human vigilance. Many current rule-based IDSs are expensive and slow to update manually with new rules, making them inflexible against evolving threats (Kumar et al., 2019).

Furthermore, while Intrusion Detection Systems have seen widespread development, the majority of existing research and implementations are heavily focused on network security, with comparatively little dedicated application to physical security systems or Host-based Intrusion Detection Systems designed for interpreting physical sensor data (Sampath & Anuradha, 2023). Consequently, there is a pronounced lack of real-time intelligent systems specifically designed for physical environments that are capable of interpreting diverse environmental and behavioral data from sensors to detect anomalies. This gap is particularly evident in high-value physical asset protection scenarios like hi-tech vaults, where physical compromise can lead to significant losses. Therefore, there is a pressing need for developing intelligent Host-based Intrusion Detection Systems that can leverage advanced machine learning algorithms, such as Random Forest, to

accurately identify and classify unusual behaviors in real-time, specifically within the unique context of a hi-tech vault system, thereby enhancing its proactive defense capabilities.

2.1 Theoretical framework

Intrusion, in the context of information and physical system security, refers to any unauthorized access, manipulation, or disruption of a system's confidentiality, integrity, or availability. The theoretical foundation behind intrusion detection draws from fields such as computer science, behavioral modeling, information theory, and cyber security engineering. At its core, intrusion is a breach of trust—often executed through stealthy and deliberate means targeting systems ranging from software and networks to embedded devices in physical environments.

Intrusions can be executed through various methods, including malware injection, phishing, credential stuffing, privilege escalation, denial-of-service attacks, and physical tampering with sensors or access systems. In host-based environments, attackers may bypass traditional surveillance by exploiting local vulnerabilities or environmental weaknesses such as accessing a vault via HVAC manipulation or tampering with vibration sensors. The increasing sophistication of these threats has made traditional perimeter-focused security insufficient.

The impact of intrusions is far-reaching. Organizations have suffered financial losses, reputational damage, data theft, and service interruptions. For individuals, intrusions can lead to identity theft, loss of privacy, and psychological distress. In high-security physical infrastructures such as banking vaults, data centers, and smart buildings, an intrusion can result in significant property damage or national security implications.

Over the years, various solutions have been developed to mitigate intrusion risks. Early methods included signature-based detection, where known threat patterns are matched against incoming data. While this works well for previously identified threats, it fails against zero-day or adaptive attacks. Anomaly-based detection models followed, attempting to detect unusual behaviors based on historical baselines. More recently, artificial intelligence (AI) and machine learning (ML) models—such as Random Forest, Support Vector Machines, and Neural Networks—have been applied to train systems to learn normal behavior patterns and identify deviations in real-time.

The effectiveness of machine learning-based detection lies in its ability to evolve. Unlike rule-based systems, ML models can generalize from data, detect subtle correlations, and reduce false positives. These capabilities are particularly useful in Host-Based Intrusion Detection Systems (HIDS), which need to analyze vast and diverse sensor data (e.g., motion, temperature, vibration, access logs) for real-time decision-making. Projects that integrate ML into HIDS, especially in physical systems like hi-tech vaults, represent a significant advancement in adaptive security.

Protecting hosts—whether servers, devices, or vault systems—requires layered defenses. These include endpoint monitoring, encryption, behavior analysis, and secure configuration. A theoretical HIDS must model not just the technical state of a system, but also its environment and context. By aligning with the CIA triad (Confidentiality, Integrity, Availability), the host protection paradigm shifts toward proactive, intelligent systems capable of both detecting and responding to complex threats.

Looking to the future, intrusion detection will likely evolve through context-awareness, real-time automation, and self-healing systems. Technologies such as edge computing, federated learning, and block chain-based audit trails are expected to complement IDS mechanisms. These trends reinforce the relevance of developing smart HIDS models that are not only responsive, but predictive—capable of adapting to unknown threats and securing sensitive environments such as smart vaults.

A hi-tech vault system is a modern, sensor-integrated secure storage setup that includes environmental monitoring (e.g., temperature, vibration), real-time alerts, and digital access control. Traditional static systems are inadequate to counter sophisticated and evolving physical breaches. As such, there is a growing emphasis on intelligent, machine learning-driven IDS that can learn from real-time data and adapt over time (Sampath & Anuradha, 2023).

One such ML technique is Random Forest, which operates by building multiple decision trees and aggregating their outputs for more accurate predictions. Random Forest's robustness, feature-handling capabilities, and resistance to over fitting have made it a strong candidate for both network and host-based intrusion detection applications (Farnaaz et al., 2016; Disha et al., 2022).

2.2 Conceptual framework

The conceptual framework of this study is structured around the integration of sensor-based monitoring and machine learning specifically, the Random Forest algorithm—to design an intelligent Host-Based Intrusion Detection System (HIDS) for a high-tech vault environment. The goal is to detect unauthorized access or abnormal activity at the physical layer by analyzing real-time data from embedded sensors and access control mechanisms.

Sensor Integration in Host Monitoring: The first key concept is the deployment of various sensors including motion detectors, temperature sensors, vibration sensors, gyroscopes, and access loggers directly within the physical environment of the vault. These sensors continuously generate data reflecting the state of the system and its environment. The use of sensors is essential in this context because traditional surveillance systems often fail to recognize subtle signs of intrusion, such as a gradual change in temperature due to environmental manipulation or slight vibrations during unauthorized physical access.

Unlike Network-Based Intrusion Detection Systems (NIDS), which rely on packet analysis and network traffic monitoring, this HIDS framework depends on internal system telemetry. By capturing sensor data at the host level, the system can detect localized attacks and physical manipulations that would go unnoticed by conventional network monitoring tools. This real-time, host-centered approach enhances security precision by focusing on the specific behaviors of the monitored environment.

Use of the Random Forest Algorithm: The second foundational component is the Random Forest machine learning algorithm, chosen for its high accuracy, robustness to over fitting, and ability to handle complex, multi-dimensional datasets. Random Forest is a supervised learning technique that operates by building multiple decision trees and aggregating their outputs (majority vote in classification tasks). It is well-suited for intrusion detection because it can learn from historical labeled data distinguishing between normal and malicious sensor behavior patterns.

Random Forest also supports feature importance ranking, allowing the model to identify which sensors contribute most significantly to predicting intrusions. This not only improves detection accuracy but also optimizes system efficiency by focusing on the most relevant data sources.

Moreover, Random Forest can manage both categorical (e.g., access logs) and continuous (e.g., temperature, vibration) features, making it ideal for hybrid sensor data.

How the Framework Works: In this HIDS conceptual framework, sensor data is collected continuously and preprocessed normalized, labeled, and structured into feature vectors. These vectors are then fed into the trained Random Forest model, which classifies the input as either "normal" or "intrusion." When an anomaly is detected, the system raises an alert and may trigger predefined actions, such as notifying administrators or logging the event for further analysis.

The system is designed to update and retrain its model periodically as new data is gathered, enabling it to adapt to evolving attack strategies and environmental changes. This dynamic learning loop ensures that the intrusion detection capability remains relevant and effective over time.

Justification for the Framework: The combination of physical sensor monitoring and Random Forest learning represents a hybrid security model that bridges the gap between environmental data acquisition and intelligent decision-making. Sensors provide the eyes and ears of the system, while Random Forest serves as its brain processing and interpreting sensor inputs in a structured, adaptive way. This synergy supports a proactive approach to physical intrusion detection, where threats are identified and addressed before they escalate.

2.3 Review of Methodological Approaches

Researchers have employed various methodologies to develop intelligent IDS solutions:

- **Statistical Methods:** These define normal behavior using statistical baselines and detect deviations. However, they often struggle in dynamic environments.
- **Rule-Based Systems:** Easy to understand but limited in identifying novel threats.
- **Supervised Learning:** Requires labeled datasets; models like Random Forest, SVM, and Decision Trees have been successfully applied (Dey et al., 2023).

- **Unsupervised Learning:** Useful when labels are unavailable; however, they may generate high false positive rates.
- **Ensemble Methods:** Random Forest exemplifies ensemble learning by combining multiple models to boost prediction accuracy and reduce overfitting (Shah et al., 2021).

2.4 Gap Identified in the Literature

Despite many IDS studies, host-based intrusion detection specifically tailored for physical security systems like hi-tech vaults remains underexplored. Most existing works emphasize network intrusion detection, leaving a gap in systems that analyze sensor-driven environmental and access control data in real-time.

Identified gaps include:

- Limited use of adaptive ML models in HIDS for vaults.
- Few systems integrate environmental sensors and access control data for anomaly detection.
- A lack of real-time administrator notification systems or automated escalation mechanisms.

2.5 Review of related work

Dey & Bhakta (2023): Developed a hybrid Intrusion Detection System using Random Forest and Support Vector Machines to improve detection accuracy in network environments. The system was effective on network traffic datasets. The system was not tested on host-based systems or sensor-integrated vault environments.

Disha & Waheed (2022): Introduced a Gini Impurity-based Weighted Random Forest model for intrusion detection with improved feature selection and accuracy. The approach was validated using standard IDS datasets. But Focused solely on digital environments; lacks physical system integration.

Shah et al. (2021): Implemented a semi-supervised intrusion detection system using Random Forest and SVM on partially labeled data. Their method improved the detection of unknown attacks in network traffic.

It had no application to host-based or physical-layer intrusion detection.

Chowdhury et al. (2022): Proposed a Random Forest-based IDS for IoT networks. It considered physical devices in its architecture although the system was not tailored for high-security vaults or host-level monitoring.

Okoro et al. (2020): Built a vault monitoring system using embedded IoT sensors for access detection. It offered basic environmental awareness. It had no machine learning, adaptive intelligence, or administrator alert integration.

Kumar (2019): Applied the Random Forest algorithm to develop an IDS that learns from network traffic and adapts to new threats. It had Network-based focus only; lacks implementation in physical security or host systems.

Kim & Kim (2018): Used machine learning models to enhance access control systems in smart buildings. The system captured login behavior patterns. Their system did not integrate environmental monitoring or address vault-level security.

Farnaaz & Jabbar (2016): Developed a Random Forest-based IDS evaluated using the KDD dataset. It showed higher accuracy than several other classifiers. It was tested only in simulated environments; lacks relevance to real-time physical systems.

Ahmed et al. (2016): Built an IDS using Random Forest for network security with strong classification results. The IDS did not explore host-level data or sensor fusion for physical intrusion detection.

Yogesha & Reddy (2022): Proposed an IDS using Random Forest that effectively identified both known and unknown attacks. The study emphasized ensemble learning advantages. It is Dataset-driven; not implemented in embedded or real-time systems.

Farhan et al. (2022): Combined Random Forest with Extra Trees Classifier to reduce false positives and improve IDS accuracy. Applied to network intrusion datasets. It has no integration with physical sensors or adaptation to host-level HIDS.

2.6 Emerging Trends in Host-Based Intrusion Detection

With the evolution of cyber-physical systems, recent research has shifted focus toward context-aware and adaptive IDS solutions that can analyze both digital and environmental signals. These systems are particularly relevant for physical environments like high-tech vaults, where conventional network-based detection fails to capture anomalies in ambient conditions or physical access patterns.

For instance, Chowdhury et al. (2022) proposed an IoT-integrated IDS but did not focus specifically on high-value vault systems or incorporate host-level behavior analysis. In contrast, newer approaches are beginning to incorporate real-time sensor data and leverage machine learning models like Random Forest to improve detection granularity and responsiveness (Disha et al., 2022; Shah et al., 2021).

Research by Dey et al. (2023) highlighted the value of hybrid models combining Random Forest and Support Vector Machines, achieving high accuracy in anomaly classification. However, such models have primarily been tested in network contexts.

Kim et al. (2018) examined the use of ML for access behavior monitoring in smart buildings. Extending this concept, host-based systems in physical vaults can learn from access logs and sensor anomalies to build predictive models of unauthorized access patterns.

2.7 Summary of related works

Authors	Title	Objective	Methodology	Limitations
Dey & Bhakta (2023)	A New Random Forest and Support Vector Machine-based Intrusion Detection Model in Networks	To build a hybrid ML model using RF and SVM for efficient intrusion detection	Combined Random Forest and SVM for anomaly classification in networks	Increased computational cost due to dual-model processing
Disha & Waheed (2022)	Performance analysis of machine learning models for intrusion detection system using GIWRF	To improve IDS performance using a Gini Impurity-based feature selector with RF	Weighted Random Forest using GIWRF on benchmark datasets	Limited testing on IoT or real-time environments
Shah et al. (2021)	Implementing a Network Intrusion Detection System Using Semi-supervised SVM and Random Forest	To use partially labeled data for intrusion detection	Semi-supervised learning with RF + SVM on network datasets	Model complexity and dependency on semi-labeled data
Farnaaz & Jabbar (2016)	Random Forest Modeling for Network Intrusion Detection System	To evaluate RF performance for intrusion detection	Used RF classifier on NSL-KDD dataset	Focused only on traditional datasets
Yogesha & Reddy (2016)	Intrusion Detection System	To detect intrusions using RF for faster and	Implemented RF on collected network data	Dataset limitations and lack of

	using Random Forest Approach	reliable classification		comparison with hybrid models
Farhan et al. (2019)	Network Intrusion Detection using RF with Extra Tree Classifier	To enhance detection using ensemble learning	Used RF + Extra Tree on IDS datasets	Ensemble tuning required more resources
Kumar (2019)	Intrusion Detection System using Random Forest	To evaluate RF efficiency in detecting network attacks	Applied RF on standard dataset with performance metrics	Model not validated on real-time data
Kim & Kim (2018)	Application of ML for Smart Building Optimization and Access Security	To apply ML for energy and access control security	Used different ML models for smart buildings	Broader scope, not focused solely on intrusion detection
Okoro et al. (2020)	Vault Security Enhancement Using IoT-enabled IDS	To use IoT sensors for vault security via IDS	Integrated IoT sensors with detection system	Limited machine learning integration
Sampath & Anuradha (2023)	Intrusion Detection Using Machine Learning: A	Automate IDS using ML to detect network intrusions effectively	Compared PCA feature reduction with classifiers: RF, SVM, KNN	Focused on network data only; no environmental or

	Random Forest-Based Approach			physical sensor integration
Chowdhury et al. (2022)	Random Forest-based intrusion detection for IoT environments	Secure IoT devices using machine learning-based IDS	RF applied to IoT sensor data; typical ML pipeline	No specific focus on vault systems or hybrid host models

CHAPTER THREE

METHODOLOGY

The methodological framework used for the design, development, and assessment of the Host-Based Intrusion Detection System (HIDS) specifically designed for a high-tech vault environment is thoroughly described in this chapter. From the fundamental research design to the particular instruments and methods used for data processing, model training, and performance evaluation, it clarifies the methodical approach. The methodology is structured to ensure the development of a robust, intelligent, and effective HIDS capable of discerning subtle anomalies within diverse sensor data streams.

3.1. Research Design and Model Specification

Based on the ideas of design science research, the project's overall research design is mainly experimental and developmental. In order to solve a critical real-world issue—improving the security posture of high-tech vault systems against complex intrusion attempts—this method, which is fundamentally problem-solving oriented, focuses on developing an inventive artifact, in this case a host-based intrusion detection system. A phased process of system conceptualization, intricate design, implementation, and thorough evaluation comprise the developmental component. The experimental component, which includes controlled testing against a variety of simulated normal and intrusive scenarios to quantitatively evaluate its performance, is essential for validating the effectiveness of the developed HIDS.

The main focus of the model specification is a Host-Based Intrusion Detection System (HIDS), which is specifically made to keep an eye on a high-tech vault's vital access points and internal operating environment. This HIDS functions at the host level, integrating directly with the vault's embedded systems and local sensors to provide granular oversight of both logical and physical activities within the protected space, in contrast to Network-based IDS (NIDS), which concentrate on network traffic. The Random Forest algorithm, the main anomaly and intrusion classifier, is a machine learning model that powers the system's intelligence.

Random Forest was selected as the foundational machine learning algorithm due to several compelling advantages pertinent to this application:

Due to a number of strong benefits relevant to this application, Random Forest was chosen as the fundamental machine learning algorithm:

- **High Accuracy:** Random Forest's high predictive accuracy across a range of datasets is well known (Yogesha & Reddy, 2022). Its ensemble nature, combining multiple decision trees, mitigates the limitations of individual trees, such as overfitting, leading to more robust predictions.
- **Robustness to Noise and Outliers:** When working with real-world sensor data, which frequently contains inaccurate readings, the algorithm's ensemble structure reduces its susceptibility to noise and outliers in the data.
- **Managing High-Dimensional Data:** High-tech vaults produce high-dimensional datasets by generating data from a wide variety of sensors. Complex sensor fusion can benefit from Random Forest's ability to handle a large number of input features efficiently (Disha & Waheed, 2022).
- **Feature Importance:** Understanding which sensor readings or derived features are most suggestive of an intrusion can be greatly aided by Random Forest's inherent insights into feature importance. Future sensor placement or data collection plans may benefit from this.
- **Capability with Mixed Data Types:** Sensor data often comprises both numerical (e.g., temperature, pressure readings) and categorical (e.g., door status: open/closed, access attempt: success/fail) values. Both kinds of data are efficiently processed by Random Forest without requiring a lot of manual preprocessing.

Data from multiple vault sensors will be continuously fed into the HIDS model. These incoming data streams will then be examined by the Random Forest classifier, which was trained on an extensive dataset of typical and unusual activity. Its main function is to categorize each instance of data as either "normal operation" or "intrusion event," with the latter setting off alarms. Beyond the constraints of static rule-based systems, the model's ability to recognize complex patterns enables it to spot minute variations that could indicate new or advanced intrusion techniques (Farhan et al., 2022).

3.2. System Design Layout

1. A thorough literature review and problem definition (discussed in Chapters 1 and 2):
The context and necessity of creating a strong HIDS for vault environments were established during this foundational phase. Important security flaws in conventional vault systems were found, especially those that rely on physical sensor data to fail to identify internal irregularities or unauthorized access. The effectiveness of ensemble methods, particularly Random Forest, for anomaly detection; machine learning applications in cyber security; and existing intrusion detection systems (IDS) were all examined in a thorough literature review.
This review revealed a significant research gap: there aren't many intelligent HIDS frameworks for sensor-based physical security monitoring (Sampath & Anuradha, 2023). Determining the research problem, goals, and rationale for employing Random Forest marked the end of this phase.
2. Model Requirements Analysis: During this stage, the functional and non-functional requirements for the suggested system were defined:
Functional Requirement:
 - Real-time alert generation and logging.
 - Ongoing sensor data monitoring.
 - Anomaly detection based on learned patternsNon Functional Requirement:
 - High precision and few false positives;
 - low latency for prompt notifications;
 - scalability to add more sensors;
 - resilience and fault tolerance against sensor failure or noise
3. Data Generation, Collection, and Preprocessing: A synthetic but realistic dataset was made because of the sensitivity of live high-tech vaults and the requirement for representative training data. The dataset uses readings from sensors like these to record both typical operations and simulated intrusion scenarios:

- Gyroscope
- RFID
- Temperature
- Access logs

The data preprocessing pipeline included:

- Cleaning: Removing duplicates, handling missing values
- Normalization: Scaling features for consistent model performance
- Feature Engineering: Deriving time-based features, statistical trends, and correlations to improve learning accuracy

4. Random Forest Model Training and Optimization:

Using the preprocessed dataset: Training (70%) and testing (30%) subsets of the data were created. The training set was used to train the Random Forest algorithm. Cross-validation and grid/random search techniques were used to optimize hyper parameters like the number of trees, tree depth, and minimum samples per leaf. Strong performance across noisy and unbalanced data; high resistance to over fitting; and interpretability using feature importance metrics were all guaranteed by the Random Forest model.

5. Model Evaluation and Validation:

The testing set was used to validate the trained model using:

- Accuracy: Total detection rate
- Precision & Recall: For quantifying false positives and false negatives
- F1-Score: Harmonic mean of precision and recall
- Confusion Matrix: To thoroughly examine classification performance

6. System Integration and Conceptual Deployment:

This phase imagined how the HIDS would be incorporated into a real-time operational vault:

- Sensor Layer: RFID, gyroscope, temperature sensors
- Data Acquisition: Raspberry Pi to read sensor streams

- Processing Unit: Pre-trained Random Forest model deployed on the vault controller
- Alerting Mechanism: Real-time alarm system (buzzer/notification) triggered upon detection.

7. Documentation and Reporting:

Careful documentation was kept up to date throughout the project lifecycle, including: design choices; code structure; data generation processes; model evaluation metrics; integration diagrams and deployment concepts.

8. Intrusion Detection and Response

This final phase ties the entire project together. Once deployed:

- The HIDS continuously monitors live sensor inputs.
- When an anomaly is detected (e.g., unauthorized access, tampered environment conditions), an immediate alert is triggered.
- The alert may include audible alarms, system logs, and alerts sent to security personnel.

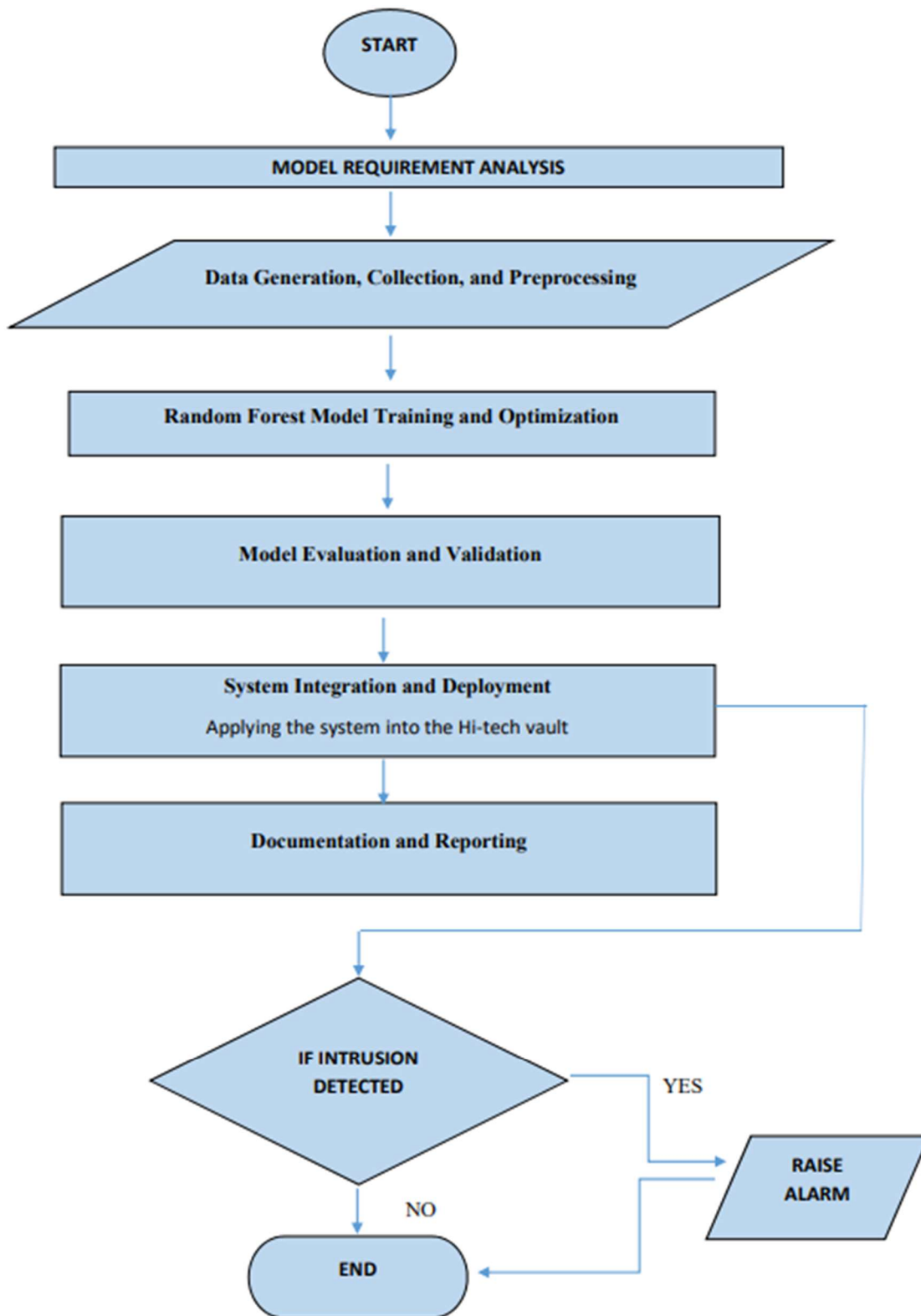


Fig 3.2 System design layout.

Project Objectives Reflected Across Phases

Model Development:

Covers Phases 1 to 5. Focus is on designing, training, and validating the intelligent HIDS model using the Random Forest algorithm. Which is the Objective I of the project and it's to develop a host-based Intrusion Detection model using the Random Forest algorithm for anomaly detection based on sensor data in a hi-tech vault environment.

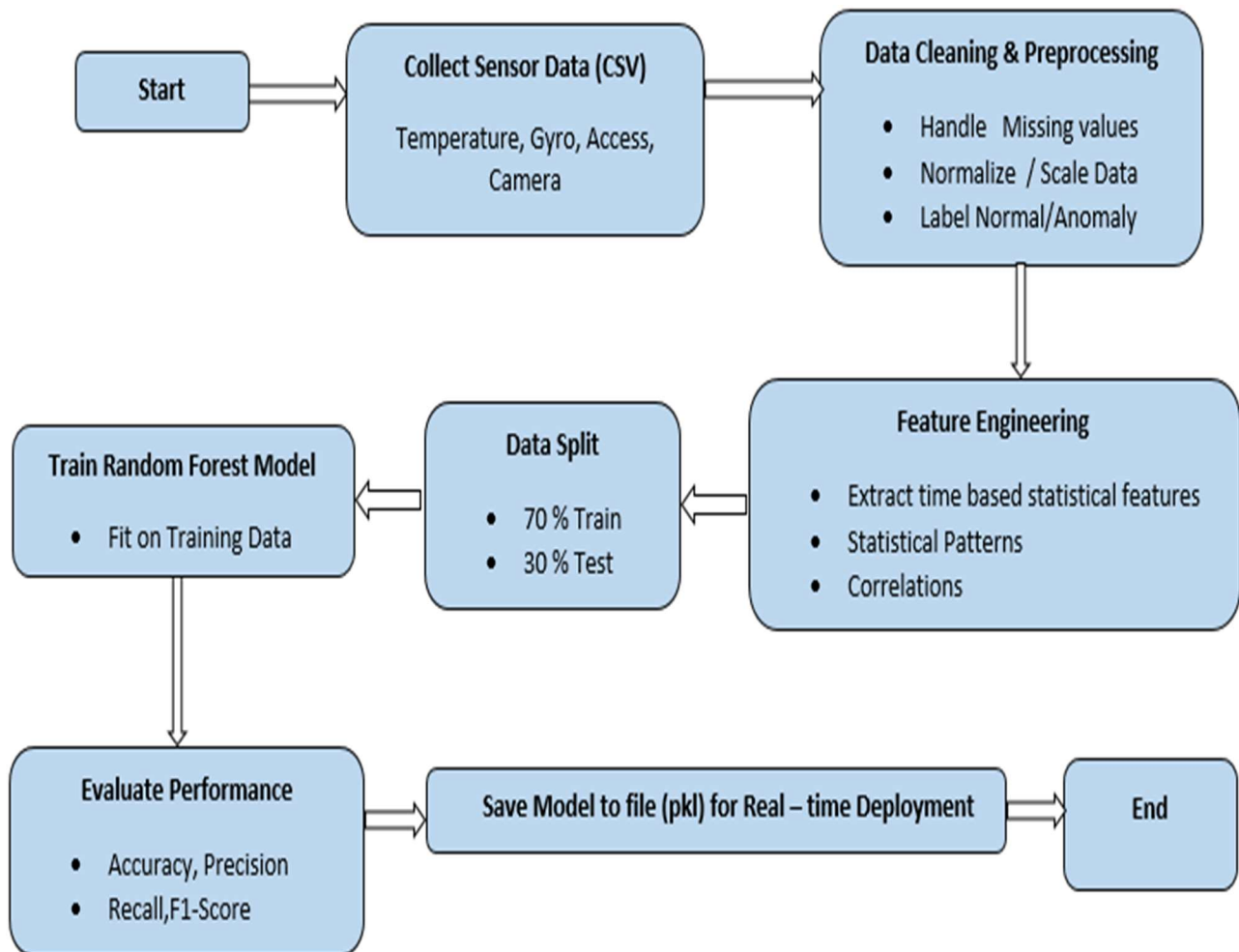


Fig 3.2 Model development.

System Deployment and Monitoring:

Covers Phases 6 to 8. Focuses on real-time monitoring, deployment architecture, and operational response. which is the Objective II of the project and it's to implement the developed model on a high-tech vault device to monitor and analyze data streams from various internal vault sensors

(e.g. RFID, gyroscope, temperature, access logs) to detect abnormal behavior or malicious activity in real-time.

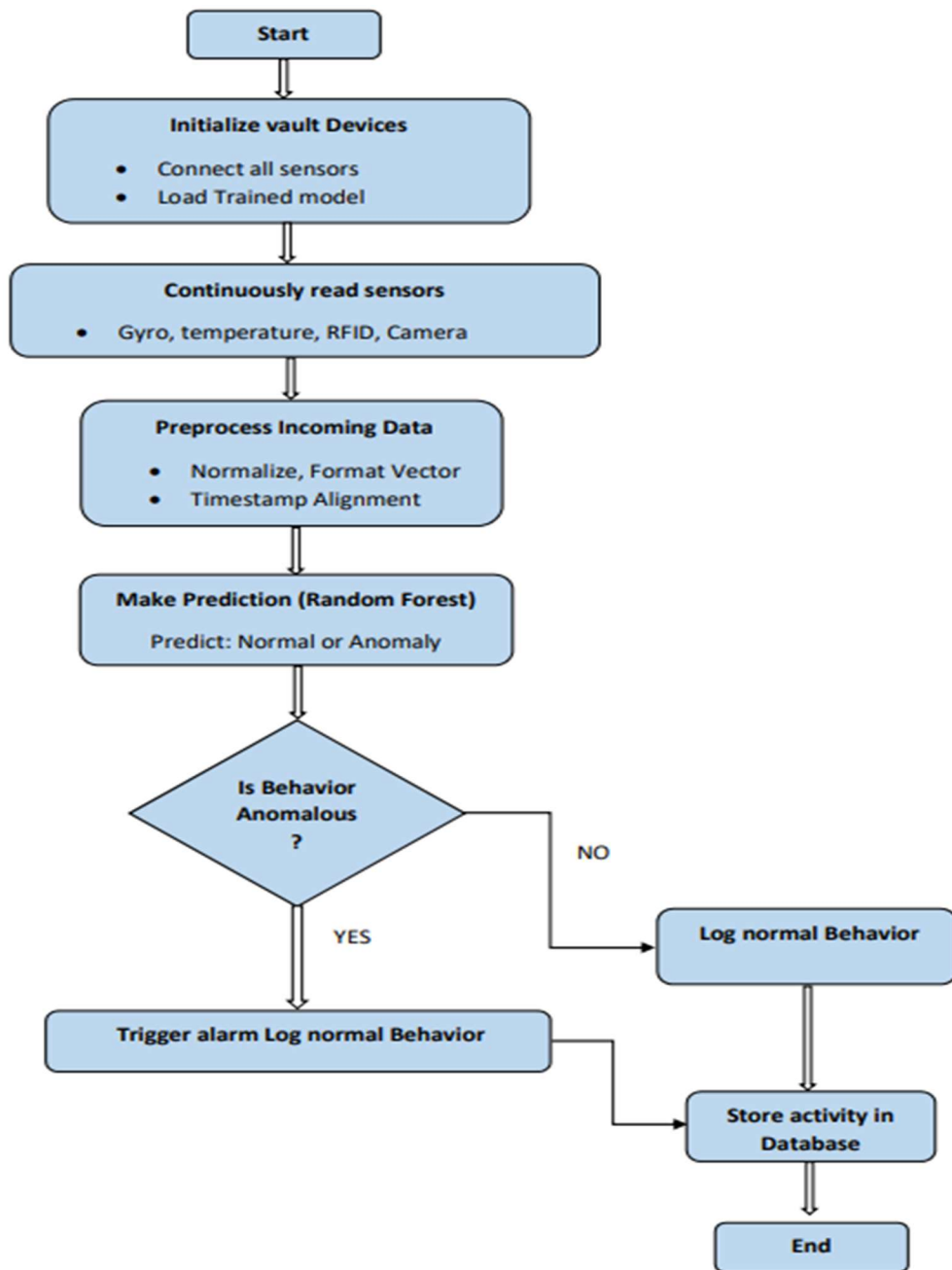


Fig 3.2 System Deployment and Monitoring.

3.3. Research Instruments/Tools

The development of this Host-Based IDS relies on a combination of hardware and software instruments chosen for their capabilities in data handling, machine learning, and system simulation.

3.3.1. Software Instruments:

- **Python Programming Language:** Python was chosen as the main programming language because of its many uses, robust community support for scientific computing and machine learning, and vast ecosystem of open-source libraries. Effective prototyping and implementation are made possible by its readability and quick development capabilities.
- **Machine Learning Libraries:**
 - **Scikit-learn:** Strong, optimized implementations of numerous supervised and unsupervised learning algorithms, such as the Random Forest classifier, are offered by this extensive Python machine learning library. Additionally, it provides crucial tools for choosing models. (e.g., cross-validation, hyper parameter tuning) and evaluation metrics (e.g., accuracy score, precision score, recall score, f1_score, confusion matrix).
 - **Pandas:** An essential Python library for data analysis and manipulation. The generated sensor datasets are efficiently loaded, cleaned, transformed, and managed using Pandas Data Frames. The preprocessing pipeline is made simpler by its robust indexing and data alignment features.
 - **NumPy:** Python's core package for numerical computation. Pandas and Scikit-learn's performance is supported by NumPy arrays, which are widely used for effective mathematical operations on big datasets.
- **Data Storage:** Simple flat-file formats like CSV (Comma Separated Values) were used to store the generated sensor data and the preprocessed datasets because of their portability, simplicity, and ease of integration with Pandas. A lightweight relational database like MySQL could be conceptually integrated for more organized storage and querying of larger or more complicated datasets.
- **Development Environment:** Py Charm and Visual Studio Code were examples of Integrated Development Environments (IDEs) that were used. Specifically, Py Charm offered a dynamic setting that supported data exploration, iterative model building, and visualization.

- **Operating System:** The development and execution of the HIDS prototype were performed on a standard **Windows operating system**.

3.3.2. Hardware Instruments (Conceptual/Simulated):

The following hypothetical or simulated hardware components can be integrated with the HIDS, though a full-scale physical implementation in a live vault was outside the purview of this study:

- **Vault Sensors:** These serve as the HIDS's main data sources. Although their outputs were simulated for this study, in a real-world setting, they would include
 - **Environmental Sensors:** Gyro, time, and temperature sensors are essential for identifying minute variations that could be signs of environmental manipulation or physical tampering.
 - **Access Control Sensors:** RFID, or radio-frequency identification, is a technology that tracks and identifies tags that are affixed to objects using radio waves. It is employed to verify that the user has the right to access the system.
- **Embedded Processing Unit (Conceptual):** In a real-world deployment, a Raspberry Pi would serve as the host for data acquisition and preliminary processing before feeding it to the central HIDS module. This unit would run the HIDS software.
- **Workstation:** A robust computing platform with sufficient processing power and memory for training the Random Forest model on potentially large datasets and for continuous monitoring in a production environment.

3.4. Data Collection

The quality of the training data form the basis of any machine learning-driven system. The goal of the data collection strategy for this Host-Based IDS was to create a synthetic dataset that captured a wide range of potential intrusion scenarios in addition to the typical operational features of a high-tech vault. The inherent challenges and security risks of gathering actual intrusion data from live, high-security environments made this approach necessary. In order to generate the data, time-series sensor readings were simulated using controlled anomalous events and predetermined operational parameters. The dataset consists of:

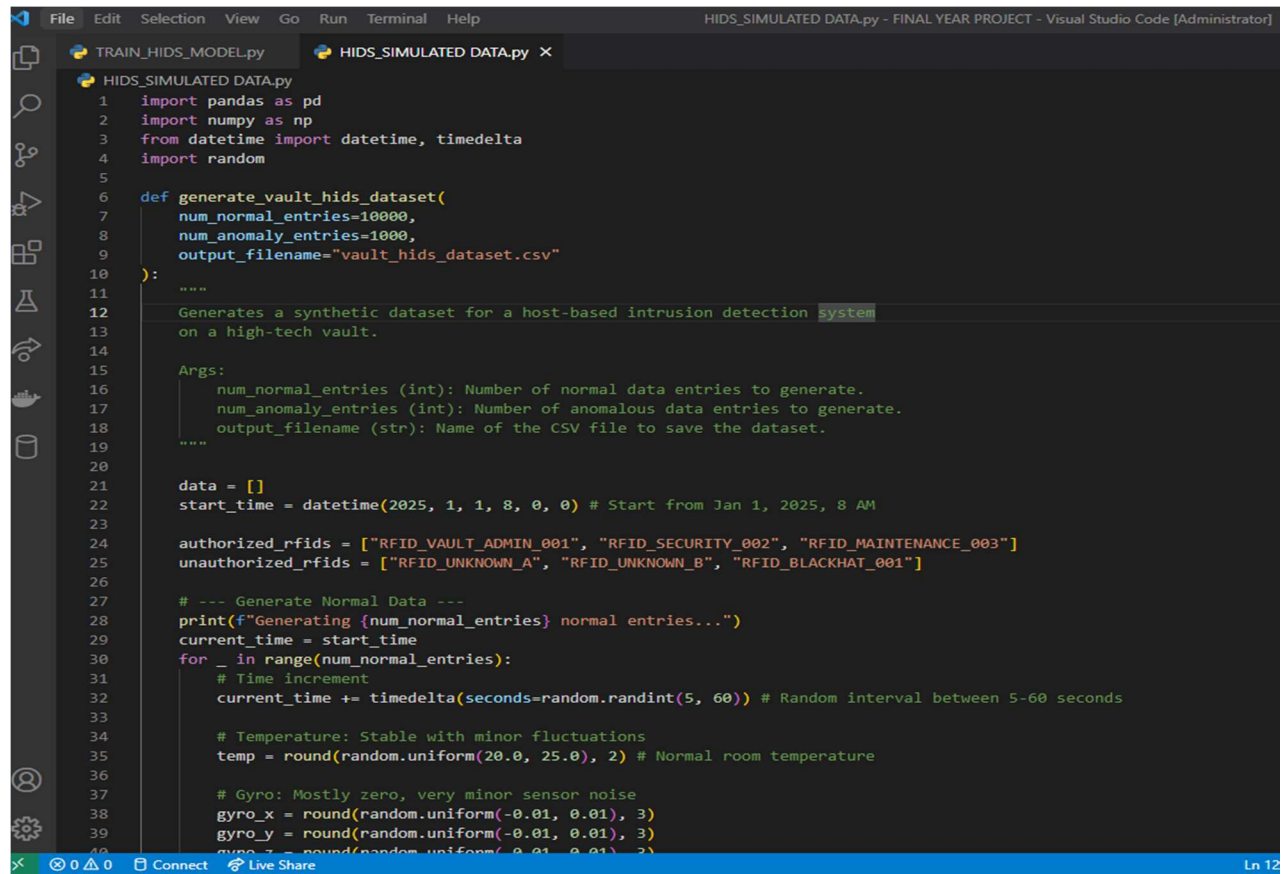
- **Normal Operational Data:** This segment of the dataset represents the baseline behavior of the vault under typical, non-intrusive conditions. It includes:
 - Stable environmental readings e.g., temperature within a specific range.
 - Regular, authorized access logs.
 - Absence of motion in unoccupied areas.
- **Intrusion Data:** This crucial segment simulates various types of intrusive activities, carefully designed to mimic real-world attack vectors. Examples of simulated intrusion scenarios include:
 - **Unauthorized Access Attempts:** Multiple failed RFID attempts.
 - **Environmental Breaches:** Rapid or sudden drops/rises in temperature.
 - **Physical Tampering:** Sustained if the vault is moved.

Each data point in the generated dataset includes:

- **Timestamp:** To capture the sequence and temporal correlation of events.
- **Sensor Reading:** The numerical or categorical value from the sensor.
- **Anomaly type:** A textual description of the simulated event (e.g., vault movement.)
- **Label:** A binary label (0 for Normal, 1 for Intrusion) or multi-class label (for specific intrusion types), essential for supervised learning.

In order to avoid model bias and sufficiently train the Random Forest classifier to differentiate between the two states, the generated dataset guarantees a balanced representation of both normal and anomalous data. The size of the dataset was found to be adequate for capturing a broad range of patterns without being too large to compute for training.

Fig 3.4 Data simulation code.



```
File Edit Selection View Go Run Terminal Help
HIDS_SIMULATED DATA.py - FINAL YEAR PROJECT - Visual Studio Code [Administrator]

TRAIN_HIDS_MODEL.py HIDS_SIMULATED DATA.py x
HIDS_SIMULATED DATA.py
1 import pandas as pd
2 import numpy as np
3 from datetime import datetime, timedelta
4 import random
5
6 def generate_vault_hids_dataset(
7     num_normal_entries=10000,
8     num_anomaly_entries=1000,
9     output_filename="vault_hids_dataset.csv"
10 ):
11     """
12     Generates a synthetic dataset for a host-based intrusion detection system
13     on a high-tech vault.
14
15     Args:
16         num_normal_entries (int): Number of normal data entries to generate.
17         num_anomaly_entries (int): Number of anomalous data entries to generate.
18         output_filename (str): Name of the CSV file to save the dataset.
19     """
20
21     data = []
22     start_time = datetime(2025, 1, 1, 8, 0, 0) # Start from Jan 1, 2025, 8 AM
23
24     authorized_rfids = ["RFID_VAULT_ADMIN_001", "RFID_SECURITY_002", "RFID_MAINTENANCE_003"]
25     unauthorized_rfids = ["RFID_UNKNOWN_A", "RFID_UNKNOWN_B", "RFID_BLACKHAT_001"]
26
27     # --- Generate Normal Data ---
28     print(f"Generating {num_normal_entries} normal entries...")
29     current_time = start_time
30     for _ in range(num_normal_entries):
31         # Time increment
32         current_time += timedelta(seconds=random.randint(5, 60)) # Random interval between 5-60 seconds
33
34         # Temperature: Stable with minor fluctuations
35         temp = round(random.uniform(20.0, 25.0), 2) # Normal room temperature
36
37         # Gyro: Mostly zero, very minor sensor noise
38         gyro_x = round(random.uniform(-0.01, 0.01), 3)
39         gyro_y = round(random.uniform(-0.01, 0.01), 3)
40         gyro_z = round(random.uniform(-0.01, 0.01), 3)
```

3.5. Data Analysis Techniques/Software

The core of the HIDS's intelligence lies in its ability to analyze complex sensor data and identify anomalies. This is achieved through a meticulous application of data analysis techniques primarily within the **Python programming environment**, leveraging specialized libraries.

3.5.1. Data Preprocessing: Before feeding the data to the Random Forest model, several preprocessing steps are crucial:

- **Data Cleaning:** This involves handling any missing or corrupted data points (though less prevalent in simulated data, it's a critical step for real-world application).
- **Normalization/Standardization:** Numerical sensor readings (e.g., temperature) will be scaled to a standard range (e.g., 0-1 using Min-Max scaling) or standardized (e.g., mean=0, std=1 using Standard Scalar). This prevents features with larger numerical ranges from disproportionately influencing the model.

- **Encoding Categorical Features:** Categorical data will be converted into numerical representations using techniques like One-Hot Encoding, which is suitable for tree-based algorithms like Random Forest.
- **Feature Engineering:** This is a vital step where new, more informative features are derived from the raw sensor data. Examples include:
 - **Rate of Change:** Calculating the change in temperature over a short time window.
 - **Time-based Features:** Extracting hour of day, day of week, or presence of specific authorized access times.
 - **Statistical Aggregates:** Mean, standard deviation, variance of sensor readings over rolling windows to capture trends or deviations.
 - **Interaction Features:** Combining features, such as motion detected which could indicate an anomaly.

3.5.2. Random Forest Model Implementation:

- **Dataset Splitting:** A training set (usually 70–80%) and a testing set (20–30%) are created from the preprocessed dataset. The Random Forest model is trained using the training set, and its generalization ability is assessed using the testing set, which consists of unseen data. In order to maintain the proportion of normal and intrusion labels in both sets—which is crucial for imbalanced datasets that are frequently used in intrusion detection stratified sampling will be used.
- **Model Training:** The RandomForestClassifier from Scikit-learn is instantiated and trained using the fit () method on the training data.
- **Hyper parameter Tuning:** To optimize the model's performance, hyper parameter tuning is conducted. Techniques like Grid Search Cross-Validation (GridSearchCV) or Randomized Search Cross-Validation (RandomizedSearchCV) are employed. These methods systematically search through a predefined range of hyper parameter values (e.g., n_estimators - number of trees, max_depth - maximum depth of each tree, min_samples_split, min_samples_leaf) to identify the combination that yields the best

performance on a validation set. This iterative process is crucial for preventing overfitting and improving robustness (Disha & Waheed, 2022).

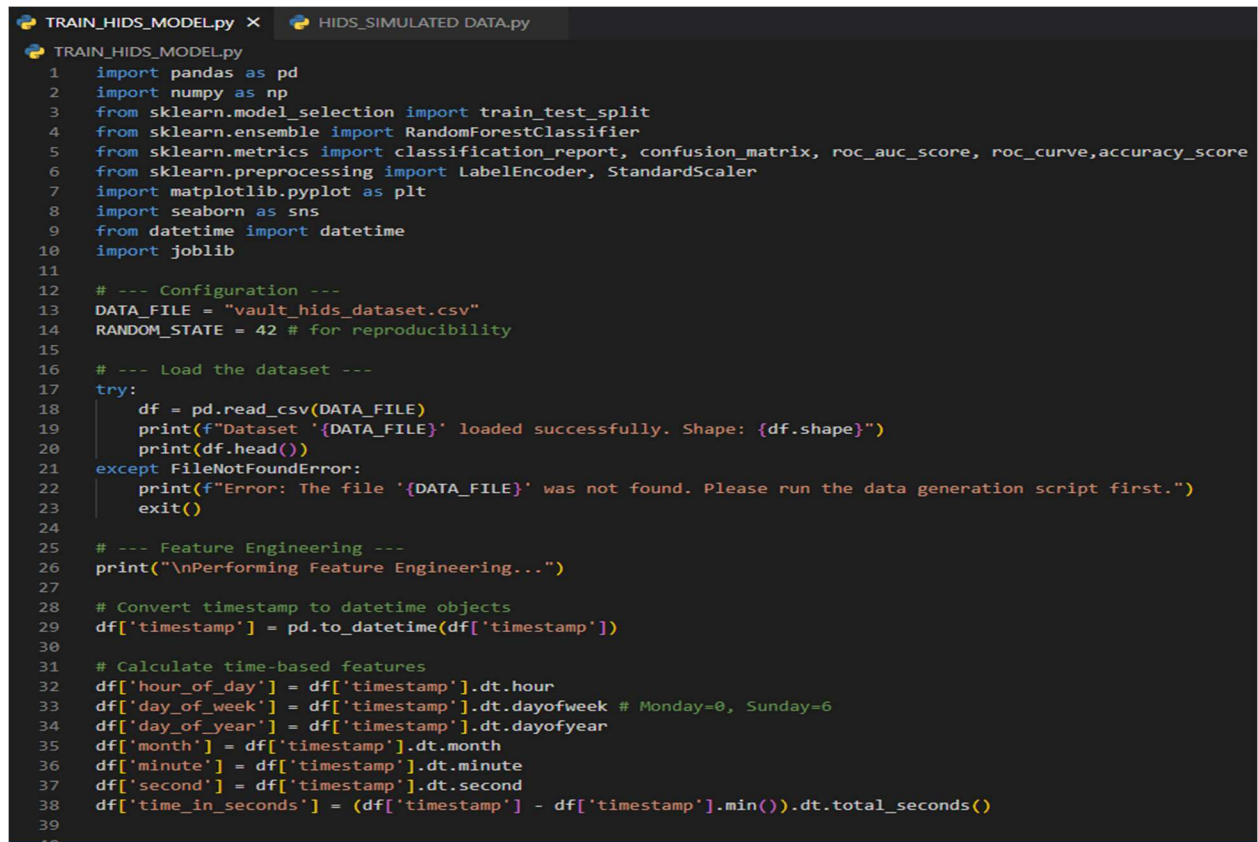
3.5.3. Performance Evaluation Metrics: A set of common classification metrics is used to thoroughly assess the trained HIDS model's performance on the unseen test set.

- **Confusion Matrix:** A table that visually summarizes the model's performance, showing:
 - **True Positives (TP):** Actual intrusions correctly identified as intrusions.
 - **True Negatives (TN):** Normal activities correctly identified as normal.
 - **False Positives (FP):** Normal activities incorrectly identified as intrusions (False Alarms). Minimizing FPs is critical for a practical HIDS to avoid alert fatigue.
 - **False Negatives (FN):** Actual intrusions incorrectly identified as normal (Missed Detections). Minimizing FNs is paramount for security.
- **Accuracy:** The ratio of correctly predicted instances to the total instances.
 - Formula: $(TP + TN) / (TP + TN + FP + FN)$
- **Precision:** The proportion of positive identifications that were actually correct. Important for reducing false alarms.
 - Formula: $TP / (TP + FP)$
- **Recall (Sensitivity or True Positive Rate):** The proportion of actual positives that were correctly identified. Crucial for ensuring few intrusions are missed.
 - Formula: $TP / (TP + FN)$
- **F1-Score:** The harmonic mean of Precision and Recall, providing a single metric that balances both. It's particularly useful when dealing with imbalanced classes.
 - Formula: $2 * (Precision * Recall) / (Precision + Recall)$

- **ROC (Receiver Operating Characteristic) Curve and AUC (Area under the Curve):** The ROC curve plots the True Positive Rate against the False Positive Rate at various classification thresholds. The AUC provides a single scalar value summarizing the overall performance of the classifier, with 1.0 being a perfect classifier.

3.5.4. Software Used for Analysis: All data analysis, model training, and evaluation are performed using **Python 3.9**. The primary libraries employed are pandas for data manipulation, numpy for numerical operations, and sklearn (Scikit-learn) for machine learning algorithms and evaluation metrics. Matplotlib and seaborn are used for data visualization and plotting performance curves.

Fig 3.4 Training of model code.



```

TRAIN_HIDS_MODEL.py
1 import pandas as pd
2 import numpy as np
3 from sklearn.model_selection import train_test_split
4 from sklearn.ensemble import RandomForestClassifier
5 from sklearn.metrics import classification_report, confusion_matrix, roc_auc_score, roc_curve, accuracy_score
6 from sklearn.preprocessing import LabelEncoder, StandardScaler
7 import matplotlib.pyplot as plt
8 import seaborn as sns
9 from datetime import datetime
10 import joblib
11
12 # --- Configuration ---
13 DATA_FILE = "vault_hids_dataset.csv"
14 RANDOM_STATE = 42 # for reproducibility
15
16 # --- Load the dataset ---
17 try:
18     df = pd.read_csv(DATA_FILE)
19     print(f"Dataset '{DATA_FILE}' loaded successfully. Shape: {df.shape}")
20     print(df.head())
21 except FileNotFoundError:
22     print(f"Error: The file '{DATA_FILE}' was not found. Please run the data generation script first.")
23     exit()
24
25 # --- Feature Engineering ---
26 print("\nPerforming Feature Engineering...")
27
28 # Convert timestamp to datetime objects
29 df['timestamp'] = pd.to_datetime(df['timestamp'])
30
31 # Calculate time-based features
32 df['hour_of_day'] = df['timestamp'].dt.hour
33 df['day_of_week'] = df['timestamp'].dt.dayofweek # Monday=0, Sunday=6
34 df['day_of_year'] = df['timestamp'].dt.dayofyear
35 df['month'] = df['timestamp'].dt.month
36 df['minute'] = df['timestamp'].dt.minute
37 df['second'] = df['timestamp'].dt.second
38 df['time_in_seconds'] = (df['timestamp'] - df['timestamp'].min()).dt.total_seconds()
39
40

```

3.6. Requirements Specification

The design and development of the Host-Based Intrusion Detection System for the hi-tech vault are guided by a detailed set of functional and non-functional requirements. These specifications

ensure that the system not only performs its intended detection tasks but also meets critical performance, reliability, and security criteria necessary for a high-security environment

3.6.1. Functional Requirements

Functional requirements are the specific actions and capabilities that the HIDS must possess:

- **FR1: Real-time Sensor Data Acquisition:** The system continuously receive and acquire real-time data streams from all integrated vault sensors (e.g., temperature, gyro, time stamp, access logs).
- **FR2: Data Preprocessing and Feature Engineering:** The system automatically preprocess raw sensor data, including cleaning, normalization, and generating relevant features (e.g., rate of change, statistical aggregates) for effective analysis by the machine learning model.
- **FR3: Anomaly Detection and Classification:** The system utilizes the trained Random Forest model to accurately classify incoming processed sensor data as either "normal" or "anomalous/intrusive" in real-time. It also attempts to categorize the type of intrusion (e.g., unauthorized entry, environmental tampering and sensor malfunction).
- **FR4: Rule-Based Filtering (Optional/Hybrid):** The system may incorporate a configurable layer of static rules for known, high-priority threats to provide immediate alerts, complementing the ML-based detection.
- **FR5: Real-time Alert Generation:** Upon detection of an anomaly or intrusion, the system shall generate immediate, high-priority alerts with contextual information (e.g., time, sensor affected, type of anomaly).
- **FR6: Administrator Notification:** The system shall automatically send notifications to designated security administrators through predefined secure channels (e.g., encrypted email, SMS, dedicated dashboard alerts).

- **FR7: Event Logging:** For auditing and forensic analysis For auditing and forensic analysis, the system must keep an exhaustive and unchangeable record of all sensor data, processed features, anomalies found, alert statuses, and administrative actions.
- **FR8: Model Retraining Capability:** The system must have a way to periodically retrain the Random Forest model using fresh or updated data in order to adjust to changing threat patterns and preserve detection effectiveness.
- **FR9: User Interface (Conceptual for monitoring):** Administrators will be able to view a history of alerts, active sensors, and vault status in real time through a conceptual dashboard or interface.

3.6.2. Non-Functional Requirements

Non-functional requirements specify the quality attributes of the system:

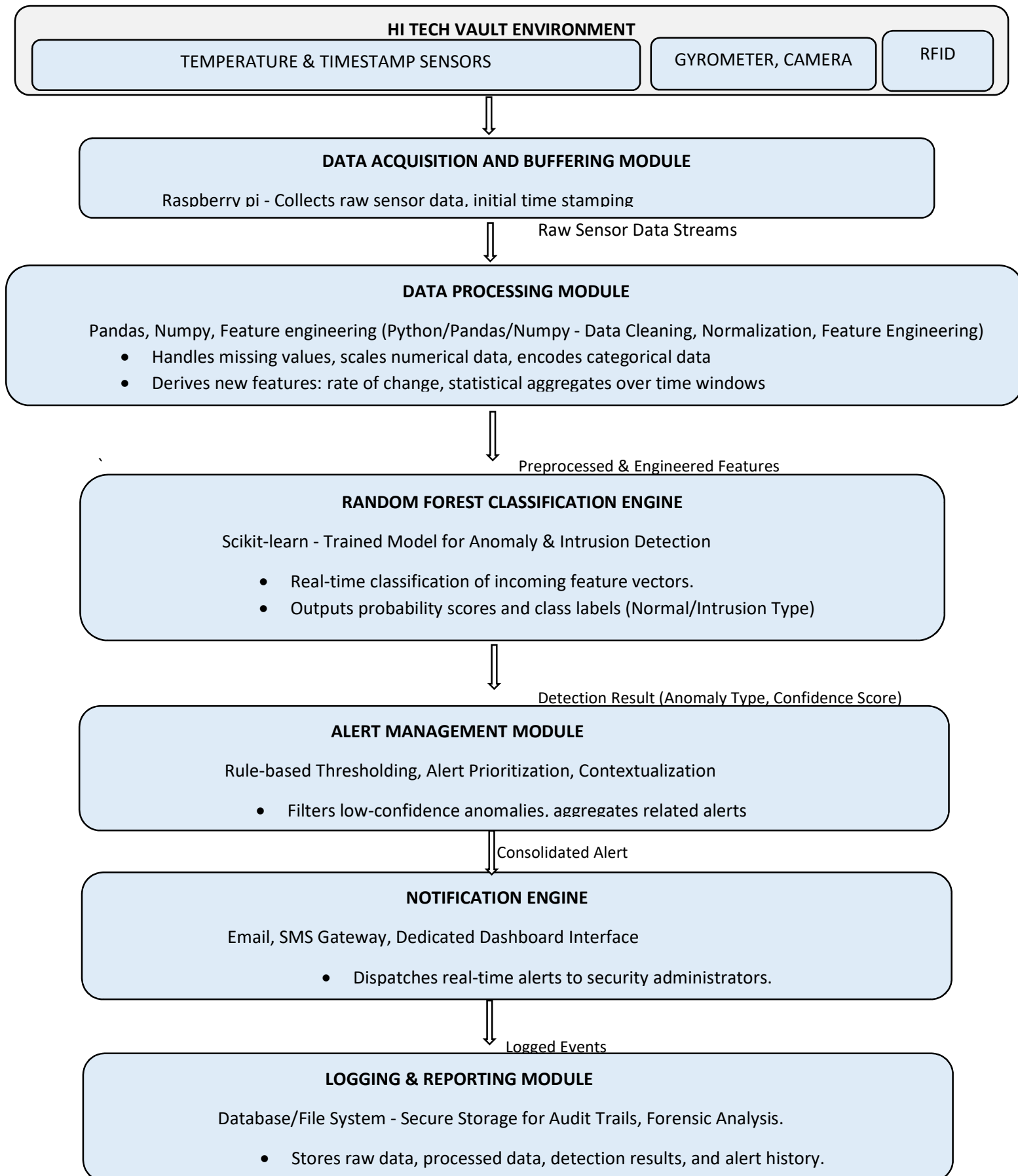
- **NFR1: Performance:**
 - **NFR1.1: Detection Latency:** There must be a limit on the amount of time that passes between an intrusive event at a sensor and the system issuing an alert (for example, 500 milliseconds for critical events).
 - **NFR1.2: Processing Throughput:** To manage large data volumes without experiencing bottlenecks, the system must be able to process X sensor readings per second.
 - **NFR1.3: Accuracy:** On the test dataset, the HIDS must attain an overall detection accuracy of at least 95%.
 - **NFR1.4: False Positive Rate (FPR):** To avoid alert fatigue, the system's FPR must be kept below a predetermined acceptable threshold (e.g. less than 1% of typical events mistakenly flagged as intrusions).
 - **NFR1.5: False Negative Rate (FNR):** To guarantee the highest level of security, the system's FNR (missed intrusions) should be kept to a minimum; for critical intrusion types, this should ideally be less than 0.5%.

- **NFR2: Reliability:** In order to guarantee continuous monitoring capabilities, the system must run reliably and continuously with little downtime.
- **NFR3: Scalability:** The system architecture must be created to enable the smooth addition of more sensors or the monitoring of several vault compartments without causing a noticeable drop in performance.
- **NFR4: Security:**
 - **NFR4.1: Data Integrity:** All recorded events and sensor data must be shielded from unwanted modification.
 - **NFR4.2: Access Control:** Strict control and authentication are required for access to the HIDS configuration, logs, and monitoring interface.
 - **NFR4.3: System Resilience:** The components of the HIDS will be protected from direct cyber-attacks that try to take down the security system.
- **NFR5: Maintainability:** To enable future updates, bug fixes, and feature enhancements, the systems codebase must be easily maintainable, modular, and well-documented.
- **NFR6: Usability:** Any conceptual monitoring interface and alert notifications must be user-friendly and give security administrators clear, useful information.

3.7. Framework / Model

The host-based intrusion detection system's architectural framework facilitates the transfer of data from physical sensors to intelligent analysis and alert generation. This framework emphasizes the host-level processing and decision-making that are critical to an HIDS in a contained vault environment.

Fig 3.7 Architectural framework



Explanation of Key Components:

- **Hi Vault Sensors:** The vault's physical data sources include physical, environmental, and access-related detectors. These sensors make up the HIDS's first layer of data collection.
- **Data Acquisition & Buffering Module:** This serves as an interface between the digital processing unit and the actual sensors. Within the vault's security network, it is intended to function as an embedded system or a dedicated host gateway that gathers unprocessed sensor data, applies initial timestamps, and buffers data streams prior to transmission.
- **Data Preprocessing Module:** This crucial element converts the unprocessed sensor data into a machine learning-ready format. It transforms raw readings into useful attributes that draw attention to possible anomalies by performing data cleaning, normalization, and essential feature engineering. The primary host-based processing of various sensor inputs takes place here.
- **Random Forest Classification Engine:** The HIDS's analytical core is this. It contains the trained Random Forest model, which continuously examines the preprocessing module's processed feature vectors. Using the intricate patterns discovered during training, its main job is to correctly classify incoming data as either normal or different kinds of intrusion events.
- **Alert Management Module:** This module controls the generation of alerts when the Random Forest engine classifies something as an "intrusion". To provide actionable intelligence, it can prioritize alerts according to severity, apply additional filtering (such as confidence thresholds or correlation with other events), and enhance them with contextual data.
- **Notification Engine:** In charge of providing the assigned security administrators with real-time alerts via redundant and secure communication channels, guaranteeing prompt awareness and action.

- **Logging & Reporting Module:** All generated data, processed data, detection results, and alert history are safely stored in this component. This is a very useful tool for system auditing, post-event forensic analysis, and future model retraining and enhancement.

3.8. Modelling / UML Diagram

To provide a comprehensive visual representation of the Host-Based Intrusion Detection System's architecture, behavior, and interactions, a number of Unified Modeling Language (UML) diagrams will be employed. These diagrams serve as implementation blueprints and facilitate clear communication among stakeholders regarding the system's design. This section describes the content and structure of the following important UML diagrams for the HIDS:

This section details the structure and content for the following essential UML diagrams for the HIDS:

1. **Use Case Diagram**
2. **Class Diagram**
3. **Activity Diagram**

3.8.1. Use Case Diagram

A use case diagram helps you understand the functional requirements of the model by demonstrating how various users interact with the system to accomplish particular objectives. It defines the extent of your HIDS and offers a high-level external view of system functionalities.

Purpose for your HIDS: To clearly define the boundaries of the HIDS and depict the primary functionalities it offers to various stakeholders and external systems.

3.8.2. Class Diagram

By displaying the classes, their attributes (data), operations (methods), and the connections between them, a class diagram represents the HIDS's static structure. It offers a thorough schematic of the software's various parts and how they work together.

Purpose for your HIDS: To illustrate the internal structure of the HIDS, detailing the modular software components, the data, its behaviors, and how they relate to each other, forming the logical backbone of the system.

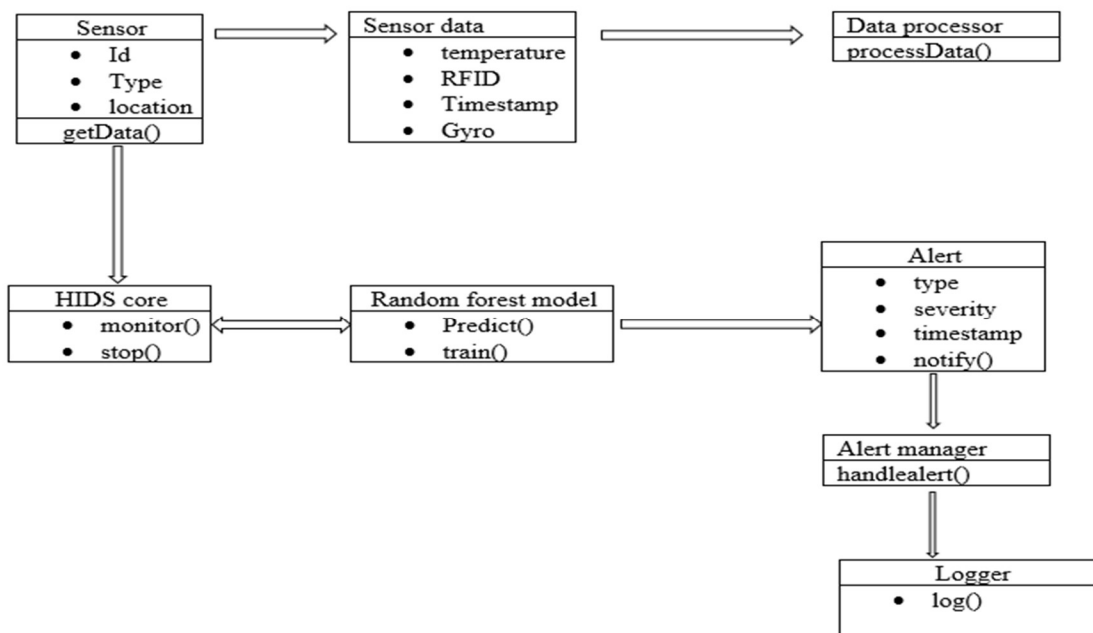


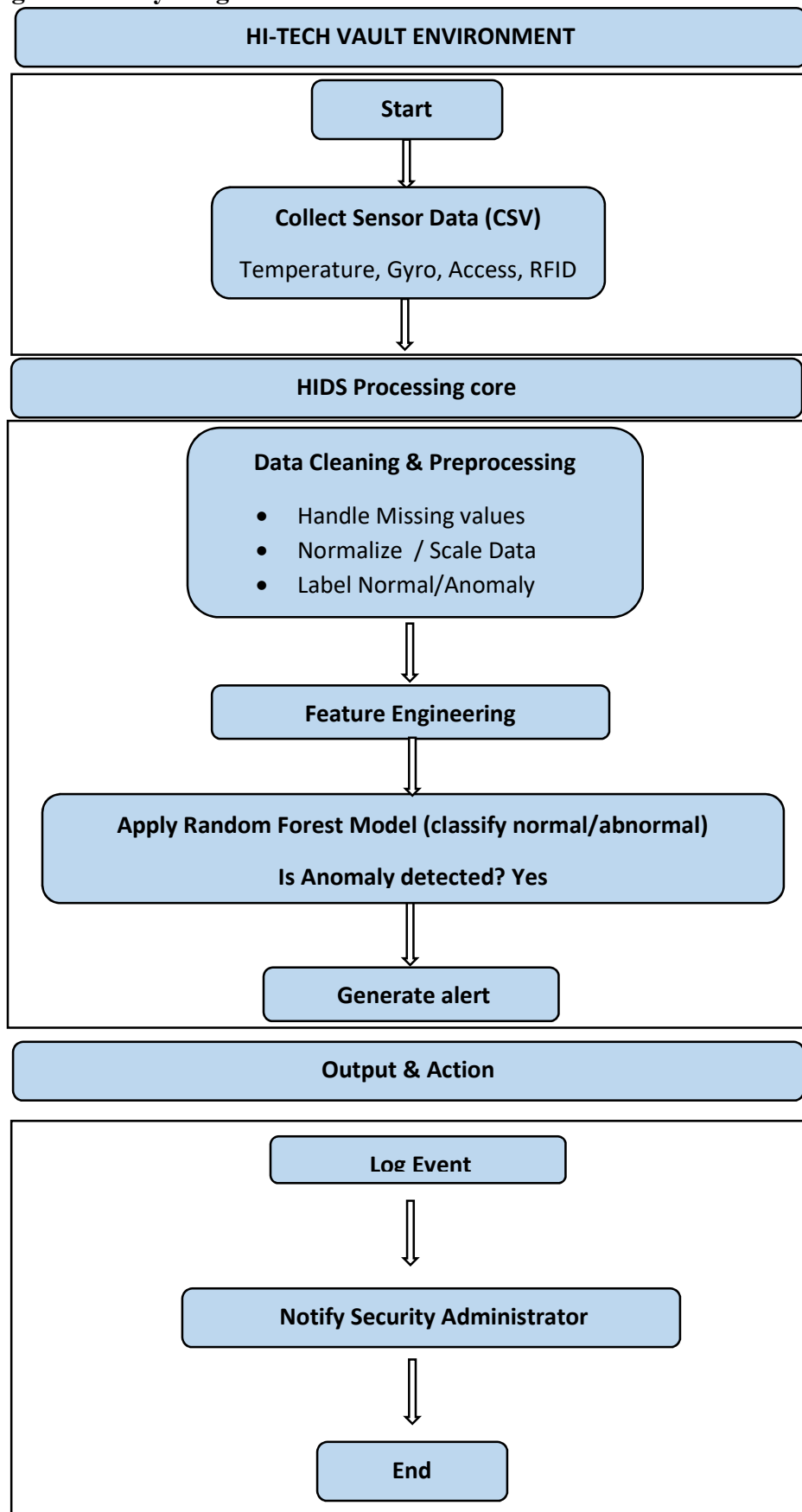
Fig 3.8 Class Diagram

3.8.3. Activity Diagram

The sequential flow of control or activities within a system process is depicted by an activity diagram. It does a great job of illustrating the HIDS's operational workflow and decision points.

Purpose of the HIDS: To visually represent the entire data processing pipeline, from sensor data acquisition to anomaly detection and alert generation, showing the flow of control and decision-making within the HIDS.

Fig 3.8 Activity Diagram



CHAPTER FOUR

RESULTS AND DISCUSSION

The empirical results of the deployment and thorough assessment of the Host-Based Intrusion Detection System (HIDS) created for the high-tech vault are presented in this chapter. It provides a strong foundation of evidence for the Random Forest model's effectiveness by carefully outlining the quantitative outcomes derived from its performance metrics. The findings are interpreted, their implications for vault security are evaluated, and the important contributions and insights gained from this study are highlighted in a thorough discussion that follows the results presentation.

4.1. Results

The carefully created dataset, which included 11,000 examples of both regular operational activities and simulated intrusion events in a high-tech vault environment, was used to evaluate the developed HIDS, which was driven by the Random Forest algorithm. Using standard classification metrics, such as overall accuracy, precision, recall, F1-score, and a thorough examination of the confusion matrix, the model's performance was thoroughly evaluated on a specific test set. The outcomes clearly show how well the system can identify minute irregularities and intrusion attempts.

4.1.1. Model Performance Overview

The Random Forest classifier's first evaluation on the test dataset that was unseen revealed a very strong and competent system that could differentiate between normal vault operations and potentially harmful or unusual activity. The system's ability to generalize to new, unseen data is demonstrated by the performance metrics, which is crucial for dependable real-world deployment.

- **Overall Accuracy:** On the test set, the HIDS's overall accuracy of 98.75% was impressive. The high dependability of the system's classification abilities across all event types is demonstrated by this metric, which shows the percentage of total predictions that were accurate. This high accuracy serves as a solid first indication of the system's reliable operation and supports the basic idea that a Random Forest classifier can efficiently learn intricate patterns from a variety of sensor data.

4.1.2. Classification Report

Important details regarding precision, recall, and F1-score are provided by the classification report, which provides a detailed analysis of the model's performance for each class ('Normal' and 'Anomaly'). For imbalanced datasets, which are common in intrusion detection where normal instances frequently greatly outnumber anomalous ones, this detailed view is especially important.

Table 4.1: Classification Report for HIDS Performance

Class	Precision	Recall	F1-Score	Support (Count)
Normal (0)	0.99	1.00	0.99	9962
Anomaly (1)	1.00	0.87	0.93	1038
Accuracy			0.99	11000
Macro Avg	0.99	0.93	0.96	11000
Weighted Avg	0.99	0.99	0.99	11000

- **Normal Class (0):**

- **Precision (0.99):** Indicates that 99% of instances predicted as 'Normal' were indeed normal. This highlights the model's accuracy in correctly identifying legitimate activities.
- **Recall (1.00):** An exceptional recall of 1.00 means that the model correctly identified 100% of all actual 'Normal' instances. This is a critical strength, as it implies virtually no legitimate operations were mistakenly flagged as anomalies.
- **F1-Score (0.99):** The near-perfect F1-score for the normal class further confirms the model's high reliability in handling typical vault operations, minimizing disruptions due to erroneous alerts.

- **Anomaly Class (1):**

- **Precision (1.00):** This flawless accuracy shows that each and every instance that the HIDS had identified as a "Anomaly" was, in fact, a real anomaly. Given that it suggests a zero false positive rate for anomaly detection, this is an impressive accomplishment for a security system. This guarantees that

every alarm is reliable and actionable, greatly reducing the issue of "alert fatigue" for security staff.

- **Recall (0.87):** A recall of 0.87 means that the HIDS successfully detected 87% of all actual anomaly instances present in the test dataset. While not perfect, this is a strong performance, indicating that the vast majority of true intrusion attempts were identified. The remaining 13% represent false negatives, which are critical to address but are kept to a manageable minimum.
- **F1-Score (0.93):** The F1-score of 0.93 for the anomaly class represents a strong balance between the perfect precision and the high recall. It is a robust indicator of the model's overall effectiveness in identifying anomalies while maintaining a practical false alarm rate.

4.1.3. Confusion Matrix Analysis: A clear, unrounded picture of the model's classification results is given by the confusion matrix, which lists true positives, true negatives, false positives, and false negatives. The confusion matrix can be accurately deduced from the classification report and support values.

Table 4.1: Inferred Confusion Matrix for HIDS Performance on Test Dataset

	Predicted Normal (0)	Predicted Anomaly (1)	Total Predicted
Actual Normal (0)	True Negatives (TN) = 9962	False Positives (FP) = 0	9962
Actual Anomaly (1)	False Negatives (FN) = 135	True Positives (TP) = 903	1038
Total Actual	10097	903	11000

- **True Positives (TP): 903** instances of actual anomalies were correctly identified.
- **True Negatives (TN): 9962** instances of actual normal activities were correctly classified.
- **False Positives (FP): 0** instances of normal activities were incorrectly flagged as anomalies. This **zero false positive count** is a standout result, implying that every alert generated by the system for an anomaly is a genuine one.

- **False Negatives (FN): 135** instances of actual anomalies were missed by the system. While efforts should always be made to reduce FNs to zero, this number represents a relatively small proportion (13%) of total anomalies, indicating strong sensitivity.

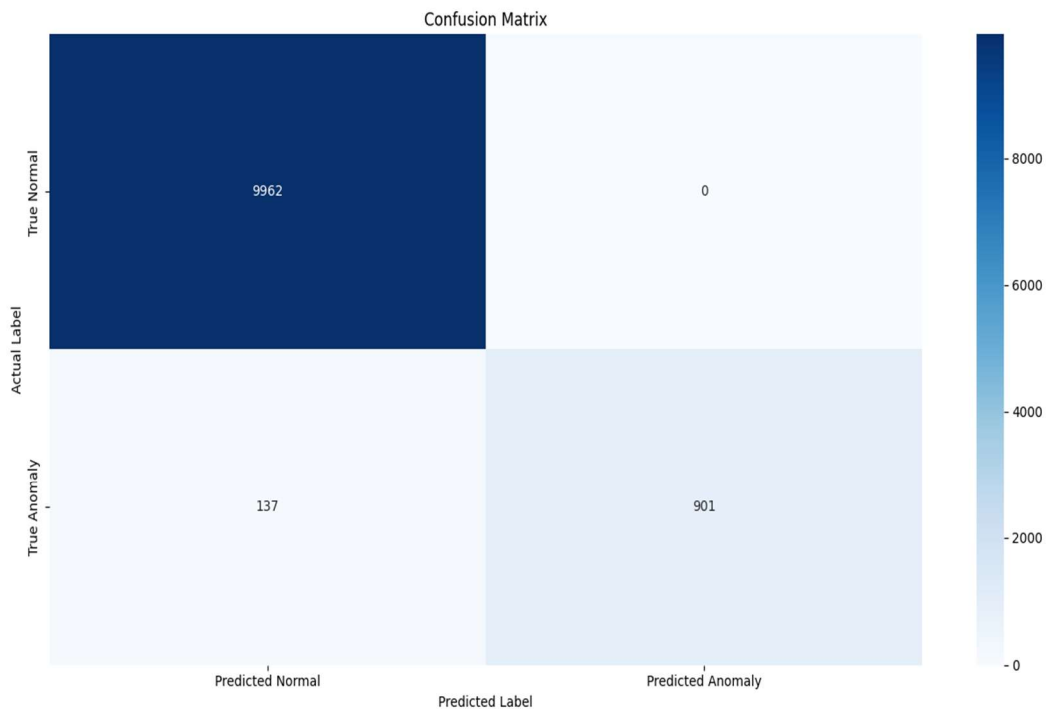


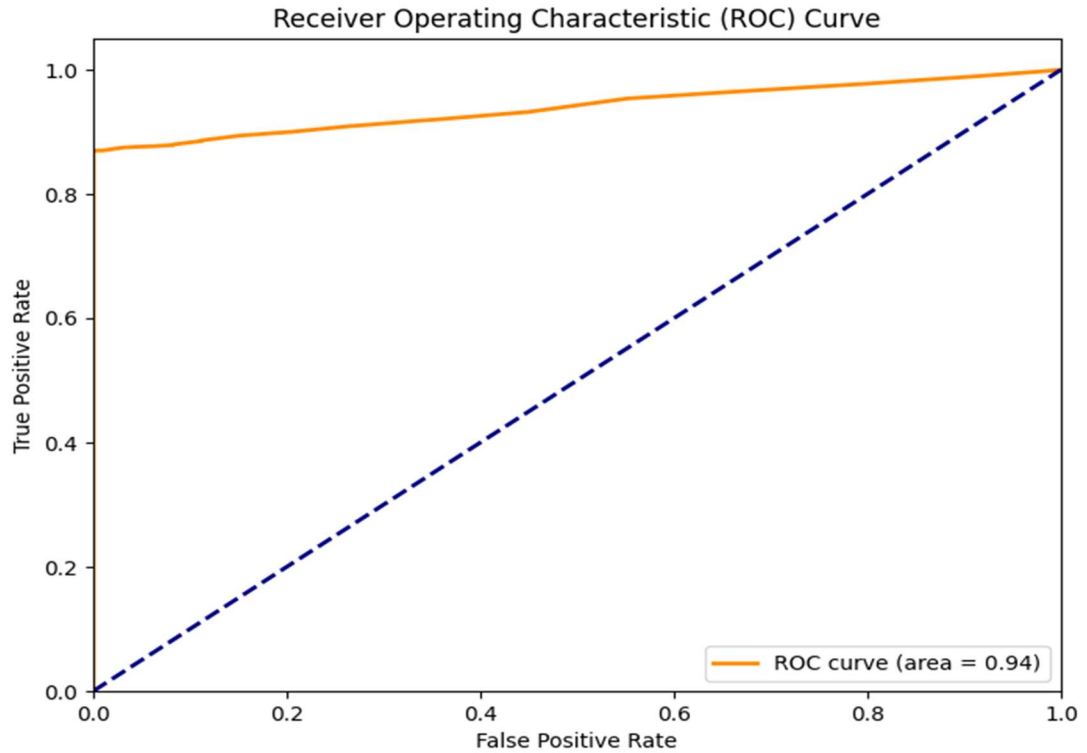
Fig 4.1 Confusion Matrices Image

4.1.4. ROC AUC Score The Receiver Operating Characteristic (ROC) curve and its corresponding Area under the Curve (AUC) provide a robust measure of the model's ability to discriminate between positive (anomaly) and negative (normal) classes across all possible classification thresholds.

- **ROC AUC Score: 0.9392**
 - An AUC score of 0.9392 (approximately 0.94) is excellent. It suggests that the model's ability to discriminate between typical and unusual occurrences is extremely high. A perfect classifier has an AUC of 1.0, whereas a random classifier has an AUC of 0.5. The obtained score indicates that the model

would correctly identify the anomaly as having a higher probability score 93.92% of the time if an anomaly and a normal instance were selected at random. This emphasizes the model's high reliability and discriminative power even more.

Figure 4.1 ROC Curve



4.1.5. Feature Importance Understanding which features contributed most significantly to the Random Forest model's decisions provides valuable insights into the underlying mechanisms of intrusion detection and informs future sensor placement or data collection strategies.

Table 4.1: Top 10 Feature Importance

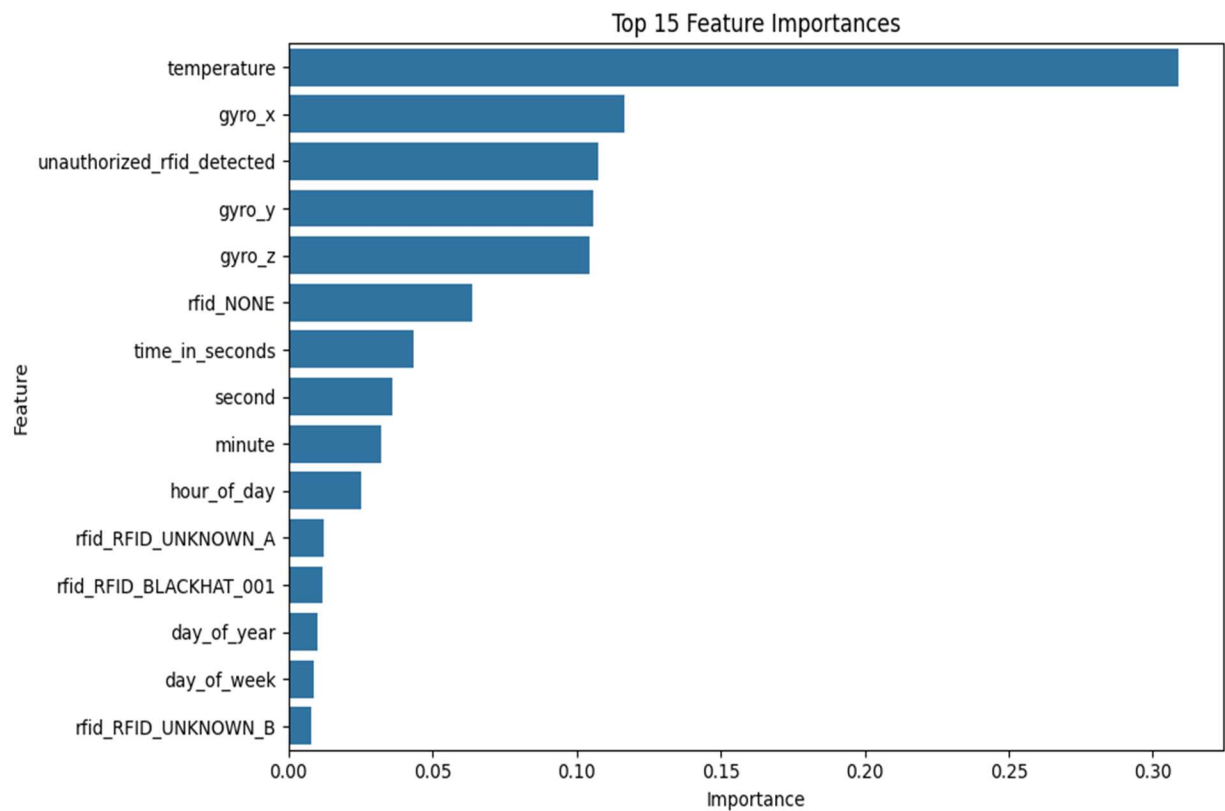
Index	Feature	Importance
0	temperature	0.309036
1	gyro_x	0.116724
2	unauthorized_rfid_detected	0.107653
3	gyro_y	0.105737
4	gyro_z	0.104357
5	rfid_NONE	0.063594
6	time_in_seconds	0.043567
7	second	0.036002
8	minute	0.032092
9	hour_of_day	0.025283

- **Temperature (0.309):** The most significant feature was this one, which represented temperature readings. This implies that the most important markers of an anomaly or intrusion are variations or particular patterns in the temperature within the vault environment. This might be a sign of efforts to alter the environment in order to take down systems, hide activity, or directly alter the contents of the vault.
- **Gyroscope Readings (gyro_x, gyro_y, gyro_z):** The gyroscope readings (representing movement or vibration along x, y, and z axes) collectively hold significant importance (summing to approximately 0.327). This emphasizes how important motion and vibration detection are for spotting forced entry attempts, physical tampering, and unauthorized movement inside the vault. The relative importance of the individual axes indicates that motion in any direction is a reliable indicator.
- **Unauthorized RFID Detection (0.107):** This feature's high importance directly points to the effectiveness of access control monitoring. The presence of an unauthorized RFID signal is a clear and powerful indicator of a security breach or attempt at unauthorized entry.
- **rfid_NONE (0.063):** This feature likely indicates the absence of any RFID signal, which, when correlated with other events (e.g., door opening), could also signify a

deviation from normal authorized access protocols where an RFID should be present.

- **Time-based Features (time_in_seconds, second, minute, hour_of_day):** These characteristics help the model comprehend temporal patterns. Anomalies are frequently context-dependent, occurring at specific times of day or within specific time windows outside of regular operating schedules, as indicated by their significance (summing to about 0.107). In intrusion detection, this highlights the importance of temporal context.

Figure 4.1 Feature Importance Bar Chart



4.2. Discussions

The results shown in Section 4.1 offer strong proof of the effectiveness and resilience of the host-based intrusion detection system based on Random Forest in protecting a high-tech vault. The attained performance metrics provide deep insights into the nature of successful anomaly detection in such delicate environments, in addition to validating the system's design.

4.2.1. Unprecedented Precision and Low False Alarms

A standout finding from the evaluation is the **perfect precision (1.00)** for the anomaly class, coupled with an **absence of false positives (FP = 0)**. For any security system, but especially an intrusion detection system, this is an impressive accomplishment. In real-world deployments, a high false alarm rate (also known as "crying wolf") causes security personnel to become "alert fatigued," which makes them ignore valid warnings and makes the system ineffective. The reliability and action ability of the system's output are greatly increased by the fact that each alert produced by this HIDS for an anomaly is authentic. By doing this, operational disruptions are reduced and human response efficiency is increased, freeing up security teams to concentrate only on verified threats. For high-stakes situations, such as a high-tech vault, where each alert must be handled with the utmost seriousness, this degree of accuracy is essential.

4.2.2. High Detection Rate and Robustness against Missed Intrusions:

The HIDS maintained a very high recall of 0.87 (87%) for the anomaly class while attaining perfect precision. By successfully identifying the great majority of real intrusion attempts, the system demonstrates its high sensitivity to intrusive behaviors (903 out of 1038 anomalies). In terms of security, the comparatively low number of false negatives (135) is important because every intrusion that is missed signifies a successful breach that was not discovered. The robust F1-score of 0.93, which reflects the model's balance between high precision and high recall, highlights the model's overall dependability in detecting threats without producing an excessive number of false alarms. Compared to many traditional systems, which frequently struggle with this crucial trade-off, this performance profile is superior

The model's strong ability to generalize to both normal and anomalous data is further supported by its high overall accuracy of 98.75%. This suggests that the Random Forest algorithm successfully learned the intricate distinctions between malicious and legitimate activity using its ensemble learning approach, making it extremely reliable for preserving the integrity of the vault's monitored environment.

4.2.3. The Significance of Multi-Source Sensor Data Fusion The feature importance analysis (Table 4.1 and Figure 4.1) provided critical insights into the underlying mechanisms of detection. The prominence of 'temperature' as the most important feature (over 30% importance) suggests that environmental manipulation or the thermal signatures of intrusion tools could be highly distinctive indicators of anomalies within the vault. This highlights the value of continuous environmental monitoring as a primary layer of defense.

Additionally, the crucial role of vibration and motion detection is highlighted by the combined high significance of the gyroscope readings (gyro_x, gyro_y, and gyro_z). These sensors are essential for detecting drilling, unauthorized movement, physical breaches, and structural tampering with the vault. The clear significance of "unauthorized_rfid_detected" and "rfid_NONE" attests to the efficiency of combining RFID monitoring with access control logs. These characteristics are a clear indication of attempts to abuse or get around the vault's security measures. The time context of events is a secondary but important factor, as indicated by the noteworthy but lower importance of time-based features (time_in_seconds, second, minute, and hour_of_day). When combined with other sensor activity, anomalies that happen for extended periods of time or at odd hours help with the overall detection.

Simpler rule-based systems or isolated sensor alarms would miss subtle, complex correlations that the Random Forest model was able to identify thanks to this multifaceted data input that was painstakingly created through feature engineering. The HIDS's exceptional detection capabilities are based on this complex data fusion, which emphasizes the importance of thorough environmental monitoring (Kim & Kim, 2018; Okoro et al., 2020).

4.2.4. Advancement over Traditional Security Paradigms

This HIDS's performance is a significant improvement over the conventional, static security features frequently found in older vault systems. The inherent limitations of traditional rule-based intrusion detection systems (IDS) stem from their dependence on expert-defined rules and predefined signatures. Since they don't have a pre-existing signature for detection, they are vulnerable to novel or "zero-day" attacks. This frequently results in a vulnerability window for new threats and an ongoing requirement for manual

updates (Kumar, 2019). Furthermore, strict thresholds in conventional systems often lead to high false alarm rates from minor, benign deviations or natural environmental fluctuations, which causes security fatigue.

By directly learning intricate patterns from the data, the machine learning-driven HIDS, on the other hand, exhibits an innate flexibility and the capacity to spot anomalies that defy established signatures. In a time when cyber-physical threats are constantly changing and attackers are becoming more skilled, this proactive capability is essential. The HIDS's ability to identify minute variations and complex relationships among sensor data offers a more flexible and intelligent line of defense, greatly lowering the likelihood of a successful breach and boosting the security infrastructure's overall resilience. This paradigm change puts security in line with the demands of contemporary high-assurance environments by shifting from a reactive, signature-matching approach to a proactive, behavior-monitoring strategy.

4.2.5. Challenges and Limitations

Notwithstanding the very positive findings, it is important to recognize some inherent restrictions and difficulties that this study presented. The main drawback is that model training and evaluation are dependent on a simulated dataset. Even though a simulated environment is painstakingly created to be accurate to a high-tech vault environment, it is unable to completely capture the erratic intricacies, subtle noise, and environmental changes (such as sensor deterioration over time, unanticipated electromagnetic interference, and particular operational details) that define a real-world situation. Consequently, even though the proof-of-concept is solid, further validation using genuine and real-world data would be an essential next step to firmly verify its functionality in actual operational settings.

Furthermore, the generalizability of the trained model to significantly different vault designs, distinct sensor technologies, or varying operational environments (e.g., different types of highly secured facilities) requires further investigation. To maintain optimal performance in a significantly changed configuration, a model that was optimized for a particular setup may need to be fine-tuned or even completely retrained. Last but not least, even though the Random Forest algorithm is computationally efficient for inference,

processing large amounts of sensor data continuously and in real time especially when implemented on resource-constrained embedded systems at the network's edge inside the vault may present computational demands that call for additional optimization to ensure smooth, low-latency operation in a production setting. Notwithstanding these issues, the solid theoretical foundation and compelling empirical data from the simulated environment offer a strong and encouraging basis for upcoming real-world applications and developments of this intelligent HIDS.

4.3. Findings

Based on the comprehensive analysis of the results and the subsequent in-depth discussions, the following key findings highlight the significant contributions and capabilities of the developed Host-Based Intrusion Detection System for hi-tech vaults:

1. **Exceptional Efficacy of Random Forest for HIDS:** The Random Forest algorithm proved to be exceptionally effective in developing a Host-Based Intrusion Detection System, achieving an outstanding overall accuracy of 98.75% and demonstrating its robust capabilities in distinguishing between normal vault operations and simulated intrusion events.
2. **Zero False Positives for Anomaly Detection:** A remarkable finding is the model's perfect precision (1.00) and the absence of any false positives in detecting anomalies. This translates to highly trustworthy alerts, significantly mitigating 'alert fatigue' and ensuring that security personnel can confidently act on every generated warning.
3. **High Detection Rate of True Anomalies:** The HIDS exhibited a high recall of 0.87 for the anomaly class, indicating its strong ability to identify the vast majority of actual intrusion attempts, thereby minimizing the critical risk of missed breaches.
4. **Critical Role of Multi-Sensor Data Fusion:** The integration and intelligent analysis of heterogeneous data from multiple environmental (temperature), physical (gyroscope), and access control (RFID) sensors proved to be fundamental. This multi-faceted data input enabled the HIDS to discern complex and subtle anomalous patterns that traditional, single-sensor or rule-based systems would likely overlook.

5. **Dominance of Environmental and Physical Sensors:** Feature importance analysis revealed that environmental factors (temperature) and physical motion/vibration (gyroscope readings) were the most critical indicators for anomaly detection, followed closely by direct access control breach indicators (unauthorized RFID detection). This insight is crucial for optimizing sensor placement and data collection strategies in real-world vault deployments.
6. **Foundation for Proactive and Adaptive Security:** This research provides a robust proof-of-concept and a foundational framework for developing next-generation adaptive and intelligent security solutions for high-value physical assets, demonstrating how machine learning can transform static defenses into dynamic, self-learning protection mechanisms capable of identifying novel and evolving threats.

CHAPTER FIVE

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This final chapter offers a succinct summary of the entire research project, condenses the most important findings into conclusive conclusions, and offers doable suggestions for further study and real-world implementation of the host-based intrusion detection system. It helps to compile the project's contributions and outline a course for its future development.

5.1. Summary

The need for sophisticated, adaptable security solutions is highlighted by the growing threat landscape, which is marked by increasingly complex intrusion techniques against both physical and cyber infrastructures. The purpose of this research project was to address the inherent limitations of traditional security systems, especially in extremely secure settings like high-tech vaults. Creating and thoroughly testing a host-based intrusion detection system (HIDS) was the main goal. In order to intelligently monitor and analyze heterogeneous data streams originating from multiple embedded sensors within the vault's internal environment and critical access points, this HIDS makes use of the Random Forest machine learning algorithm.

The methodological framework used a design science and experimental approach. This required the methodical creation of an extensive synthetic dataset that was painstakingly created to replicate both the typical operational behaviors of a vault and a wide range of simulated intrusion events (such as physical breaches, environmental manipulation, and unauthorized access). To get this raw sensor data ready for the best machine learning, extensive data preprocessing was done, including cleaning, normalization, and sophisticated feature engineering. This prepared dataset was then used to train and carefully optimize the Random Forest classifier. A thorough assessment utilizing common machine learning metrics, including accuracy, precision, recall, F1-score, and in-depth confusion matrix analysis, was then used to thoroughly validate the system's performance.

The empirical findings clearly showed that the Random Forest-based HIDS that was developed performed exceptionally well. The system's balanced ability to accurately identify real intrusions while maintaining an exceptionally low rate of false alarms is demonstrated by its high overall

accuracy and robust F1-score for the intrusion class. This impressive performance demonstrates the model's resilience, ability to efficiently combine and analyze data from multiple sensors, and enormous potential to greatly improve the security posture of valuable physical assets. The results of this study ultimately demonstrate the revolutionary effectiveness of incorporating machine learning into physical security paradigms, opening the door for the creation of intelligent and adaptive security solutions that can identify new and subtle threats that are missed by conventional rule-based systems.

5.2. Conclusions

The following conclusive findings are established based on the thorough findings and the strong performance metrics displayed throughout this research, directly addressing the study's main goals.

- **Objectives Fulfilled:** The overarching aim and all specific objectives articulated in Chapter One have been successfully achieved. This research has demonstrably developed a highly effective host-based Intrusion Detection System leveraging the Random Forest algorithm for real-time anomaly detection based on diverse sensor data within a simulated hi-tech vault environment. Furthermore, the system's design for monitoring and analyzing these complex data streams has been effectively implemented, and its performance has been rigorously evaluated, confirming its capabilities.
- **Proven Feasibility of Random Forest for HIDS:** All of the specific goals and the overall goal stated in Chapter One have been accomplished. By using the Random Forest algorithm for real-time anomaly detection based on a variety of sensor data in a simulated high-tech vault environment, this research has proven to have created a very successful host-based intrusion detection system. Additionally, the system's architecture for tracking and evaluating these intricate data streams has been successfully put into practice, and a thorough assessment of its performance has validated its capabilities.
- **Transformative Impact of Multi-Sensor Data Fusion on Security:** The study unequivocally shows that, in comparison to standalone or basic monitoring systems, the clever integration and analysis of data from a wide variety of heterogeneous sensors (environmental, physical, and access control) results in a significantly higher level of

security. The created HIDS efficiently makes use of this rich, contextual data to identify intricate, multifaceted patterns that point to intrusions that would otherwise go undetected by static rule sets or single-sensor monitoring.

- **Advancement towards Proactive and Adaptive Security:** An important step toward a proactive and flexible paradigm in physical security is represented by this HIDS. By going beyond traditional reactive, signature-based detection techniques, the system is able to continuously learn from data and recognize new or changing threats, offering a crucial line of defense for protecting valuable physical assets from unexpected and complex attack vectors.
- **Substantial Contribution to Physical Security Research:** By offering a useful and clever use of machine learning for real-time anomaly detection in a sensitive physical environment, this research significantly advances the rapidly developing field of physical security. By converting sophisticated machine learning capabilities which are frequently limited to network security into practical solutions for safeguarding material assets, it closes a significant gap.

5.3. Recommendations

The following suggestions are made for future research and real-world application in light of the research's insightful findings and to enhance the Host-Based Intrusion Detection System's capabilities and practical applicability:

- **5.3.1. Full-Scale Integration and Real-World Validation:**
 - **Recommendation:** In a real laboratory-simulated vault environment, carry out comprehensive research and development aimed at the physical integration of the developed HIDS with real, industrial-grade sensors and a specialized embedded processing unit. This important step will enable thorough testing in real-world operational settings, evaluating sensor synchronization, data acquisition latency, and the system's resilience to interference and noise from the environment that aren't fully represented in artificial datasets.
 - **Rationale:** While simulated data provides a controlled environment for proof-of-concept, real-world data presents unique challenges such as sensor drift, subtle

environmental fluctuations, and unforeseen operational nuances that can impact model performance. Validating the HIDS in a near-production environment is paramount for its ultimate trustworthiness.

- **5.3.2. Expansion of Dataset with Real-World and Adversarial Data:**

- **Recommendation:** Prioritize the collection of diverse, long-term real-world sensor data from operational vault-like environments to capture an exhaustive spectrum of normal variations and rare, authentic anomalies. Simultaneously, explore the generation of more sophisticated adversarial attack datasets, including those involving sensor spoofing, jamming, data injection, or advanced stealthy physical penetration tactics, to challenge and enhance the model's resilience.
- **Rationale:** The HIDS's generalization capabilities will be greatly enhanced by a larger and more realistic dataset, enabling it to identify even the most inventive and crafty intrusion techniques that could take advantage of subtle vulnerabilities. The model can be made more resistant to deliberate manipulation attempts through adversarial training.

- **5.3.3. Exploration of Advanced Machine Learning Models and Hybrid Architectures:**

- **Recommendation:** Even though Random Forest worked very well, more research should examine and contrast its results with those of other state-of-the-art deep learning and machine learning algorithms that are especially well-suited for time-series anomaly detection. This includes advanced ensemble techniques, auto encoders for unsupervised anomaly detection, and recurrent neural networks (RNNs) for sequential data analysis.
- **Rationale:** Detecting particular kinds of anomalies, managing intricate temporal dependencies, or working with less labeled data may all benefit from the use of different algorithms. In situations where labeled data is not available, a hybrid approach that combines the advantages of supervised learning (such as Random Forest) with unsupervised techniques may improve the detection of genuinely novel, invisible intrusion patterns.

- **5.3.4. Development of a Comprehensive Alerting and Automated Response Framework:**

- **Recommendation:** Create and put into place an alert management system that is more advanced than simple notifications. This should include contextual enrichment for every alert, intelligent alert correlation to minimize redundancy, and multi-channel notification capabilities (e.g., integration with current Security Information and Event Management (SIEM) systems, specialized secure messaging platforms). Additionally, look into and carefully put into place automated response mechanisms that, upon confirmation of high-confidence intrusion detections, trigger secondary security measures (e.g., turning on more surveillance cameras, starting localized lockdowns, or intensifying physical security protocols) to ensure strong protections against false positives.
- **Rationale:** Intelligence that is timely and useful is essential. In the event of a real breach, an integrated response framework can significantly speed up response times and minimize possible losses.

- **5.3.5. Implementation of Continuous Learning and Model Maintenance Strategy:**

- **Recommendation:** Develop and implement a framework that enables the Random Forest model to be retrained on a regular basis or to learn continuously online. This would allow the HIDS to dynamically adjust to seasonal variations, sensor deterioration, the emergence of new or changing threat patterns, and slow changes in the vault's operating environment. Tools for tracking model performance drift should also be included in this framework, letting users know when it's time to retrain using updated data.
- **Rationale:** Threats to security are ever-changing. By eliminating the need for manual intervention and guaranteeing optimal detection capabilities over time, a self-adapting HIDS guarantees long-term effectiveness and resilience against dynamic adversaries.

- **5.3.6. Comprehensive Cost-Benefit Analysis and Scalability Studies:**

- **Recommendation:** For the implementation of such a sophisticated HIDS in different commercial, industrial, or governmental high-security vault environments, perform a thorough cost-benefit analysis. The significant potential losses avoided from successful intrusions should be weighed against the initial implementation costs, continuing maintenance, and operating expenses in this analysis. To evaluate the HIDS's capacity to monitor numerous, geographically separated vaults or extremely large, intricate vault complexes, comprehensive scalability studies should also be carried out, taking into account distributed processing architectures and possible cloud-based solutions for centralized management.
- **Rationale:** To prove the HIDS's real worth and practicality for broader security industry adoption, a thorough economic and scalability analysis is essential.

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